

Numerical Simulation and Design of an Electrostatic Lens for Electron Beam Focusing

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Contents

1	Introduction	3
2	Objectives	4
3	Physical Model	4
3.1	Lens Geometry	5
4	Theoretical Background	5
4.1	Electrostatic Potential	5
4.2	Governing Equation	6
4.3	Laplace Equation in Cylindrical Coordinates	7
4.4	Boundary Conditions	7
5	Problem Formulation	8
5.1	Computational Domain	8
5.2	Modeling Assumptions	8
5.3	Simulation Parameters	8
6	Numerical Implementation	10
6.1	Grid Generation	10
6.2	Finite Difference Discretization	11
6.3	Update Equation	12
6.4	Iterative Solution Using SOR	12
6.5	Electric Field Calculation	13
6.6	Particle Trajectory Calculation	13
6.7	Beam Envelope and Focal Point Determination	14
7	Software Architecture	15
7.1	Program Structure	15
7.2	Overall Computational Procedure	15
7.3	Numerical Techniques	16
7.4	Simulation Workflow	16
7.5	Generated Outputs	17

8	Results	17
8.1	Computational Geometry	17
8.2	Electric Potential Distribution	18
8.3	Equipotential Lines	19
8.4	Electric Field Distribution	19
8.5	Electric Field Streamlines	20
8.6	Electron Trajectories	21
8.7	Beam Envelope	22
8.8	Overall Performance of the Electrostatic Lens	23
9	Parametric Study	23
9.1	Effect of Central Electrode Voltage	23
9.2	Effect of Central Electrode Voltage	24
9.3	Effect of Electrode Radius	28
9.4	Effect of Electrode Length	30
9.5	Effect of Gap	33
9.6	Effect of Initial Electron Energy	35
10	Discussion	37
10.1	Interpretation of the Results	37
10.2	Sources of Numerical Error and Model Limitations	37
10.3	Possible Improvements	38
10.4	Practical Applications	38
11	Optimization of Lens Parameters	38
12	Conclusion	40

1 Introduction

Electron beams play a fundamental role in a wide range of scientific and industrial instruments. Devices such as scanning electron microscopes (SEM), transmission electron microscopes (TEM), electron beam lithography systems, mass spectrometers, and many particle accelerators rely on precise manipulation of electron trajectories through electric and magnetic fields. The overall performance of these systems strongly depends on their ability to accurately focus and control electron beams.

Among the various beam-focusing techniques, electrostatic lenses provide an efficient method for steering charged particles without the use of optical materials. By generating appropriate electric field distributions, these lenses modify the trajectories of electrons and concentrate them into a desired focal region. One of the most widely used electrostatic lenses is the Einzel lens, which offers a relatively simple structure, effective focusing capability, and the ability to preserve the kinetic energy of charged particles throughout the focusing process.

Despite its simple geometry, the electric potential and electric field inside an Einzel lens exhibit highly non-uniform distributions. Analytical solutions of the governing equations are generally unavailable for practical geometries, making numerical methods indispensable for analyzing and designing such systems.

In this project, the electric potential inside an electrostatic Einzel lens is obtained by numerically solving Laplace's equation using the Finite Difference Method (FDM). The electric field distribution is then calculated from the resulting potential, and electron trajectories are determined through numerical integration of the equations of motion. Finally, the effects of several geometrical and electrical parameters, including the central electrode voltage, electrode spacing, and the initial electron energy, are investigated to evaluate their influence on the focusing performance of the lens.

2 Objectives

The primary objective of this project is to numerically model and analyze an electrostatic Einzel lens for electron beam focusing. To achieve this goal, the electric potential distribution inside the lens is obtained by solving Laplace's equation using the Finite Difference Method. The corresponding electric field is then calculated from the potential distribution, and the trajectories of electrons are simulated through numerical integration of the equations of motion.

Furthermore, the influence of several geometrical and electrical parameters, including the voltage of the central electrode, the spacing between electrodes, and the initial electron energy, is investigated to evaluate their effects on the focusing performance of the lens. Finally, the simulation results are analyzed to assess the overall performance of the lens and identify suitable operating conditions.

3 Physical Model

In this study, an electrostatic Einzel lens is considered for focusing an electron beam. The lens consists of three coaxial cylindrical electrodes aligned along the direction of electron propagation. The two outer electrodes are maintained at the same electric potential, typically ground potential, while the central electrode is biased at a different voltage to generate the electric field required for beam focusing.

The system is assumed to possess axial symmetry; therefore, the problem can be modeled in a two-dimensional r - z plane. In addition, the space inside the lens is assumed to be a perfect vacuum, and space-charge effects arising from interactions between electrons are neglected. Consequently, the motion of electrons is influenced solely by the electric field generated by the electrode potentials.

Since the electron velocity is assumed to be much smaller than the speed of light, relativistic effects are neglected and classical mechanics is employed to describe particle motion. Furthermore, external magnetic fields, collisions with residual gas molecules, and quantum mechanical effects are not considered in the present model.

These assumptions allow the study to focus on the fundamental behavior of the electrostatic lens while investigating the influence of geometrical and electrical parameters on its focusing performance.

3.1 Lens Geometry

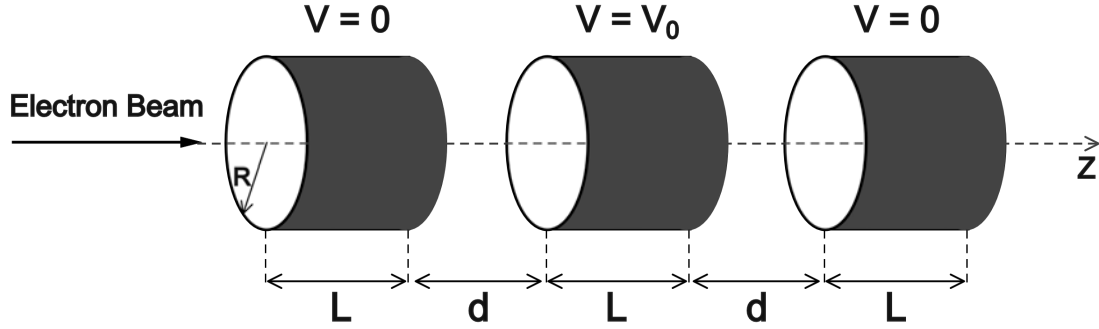


Figure 1: Three-dimensional schematic representation of the electrostatic Einzel lens consisting of three coaxial cylindrical electrodes. The two outer electrodes are grounded, while the central electrode is biased at a positive potential to produce the focusing electric field.

Figure 1 illustrates the overall geometry of the electrostatic Einzel lens used throughout this work. The cylindrical electrodes are aligned along the beam axis and generate an axisymmetric electric field suitable for electron focusing.

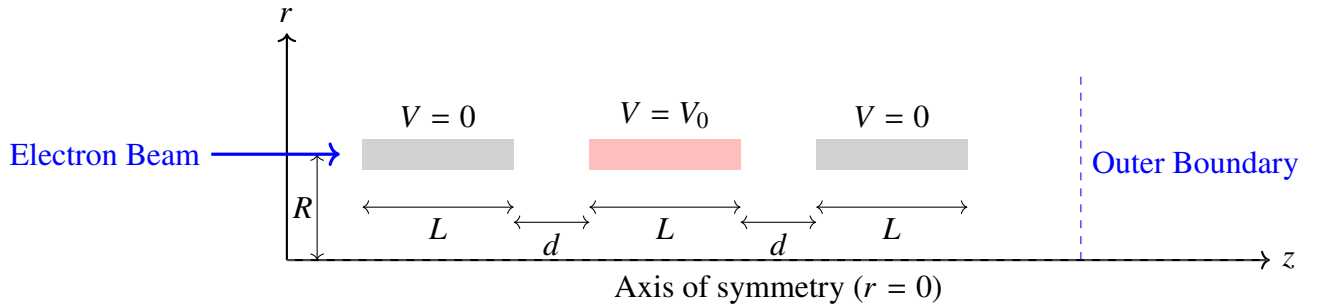


Figure 2: Axisymmetric computational model of the electrostatic Einzel lens.

Figure 1. Schematic representation of the electrostatic Einzel lens showing the electrode geometry and applied potentials.

4 Theoretical Background

4.1 Electrostatic Potential

The analysis of an electrostatic lens begins with determining the electric potential distribution inside the lens. Since the electric field can be obtained directly from the spatial variation of the electric potential, calculating the potential distribution constitutes the first step in modeling the focusing process.

In electrostatics, the electric potential is a scalar quantity representing the electric potential energy per unit charge at each point in space. For practical geometries such as electrostatic lenses, directly calculating the electric field is often difficult. Instead, the electric potential is first determined, and the electric field is subsequently derived from it.

In the present problem, the applied voltages on the electrodes create a non-uniform potential distribution within the lens. This potential distribution governs the electric field responsible for deflecting and focusing the electron beam. Consequently, obtaining an accurate potential distribution is the primary objective of the electrostatic analysis.

4.2 Governing Equation

To determine the electric potential distribution inside the electrostatic lens, the governing equation must first be derived from Maxwell's equations under electrostatic conditions.

In electrostatics, the electric field is time-independent and electric charges remain stationary. Under these conditions, Gauss's law is expressed as

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \quad (1)$$

where $\mathbf{E}$ denotes the electric field, ρ is the volume charge density, and ε_0 represents the permittivity of free space.

Furthermore, the electric field is conservative under electrostatic conditions. Consequently, its curl vanishes,

$$\nabla \times \mathbf{E} = 0 \quad (2)$$

which allows the electric field to be expressed as the negative gradient of the electric potential,

$$\mathbf{E} = -\nabla V \quad (3)$$

Substituting Eq. (3) into Gauss's law yields

$$\nabla \cdot (-\nabla V) = \frac{\rho}{\varepsilon_0}. \quad (4)$$

Using the definition of the Laplacian operator, the above equation becomes

$$\nabla^2 V = -\frac{\rho}{\varepsilon_0} \quad (5)$$

which is known as Poisson's equation.

In the present study, the region between the electrodes is assumed to be vacuum and free of space charges. Therefore,

$$\rho = 0 \quad (6)$$

and Poisson's equation reduces to Laplace's equation,

$$\nabla^2 V = 0. \quad (7)$$

Laplace's equation is the fundamental equation governing the electric potential distribution in the present numerical simulation.

4.3 Laplace Equation in Cylindrical Coordinates

As discussed in the previous section, Laplace's equation governs the electric potential distribution inside the electrostatic lens. Before solving this equation numerically, an appropriate coordinate system must be selected.

Since the Einzel lens consists of three coaxial cylindrical electrodes, the problem possesses axial symmetry. Therefore, cylindrical coordinates (r, θ, z) provide the most suitable framework for modeling the system.

Furthermore, due to the rotational symmetry of the lens, the electric potential is independent of the azimuthal angle. Consequently,

$$\frac{\partial V}{\partial \theta} = 0. \quad (8)$$

The general form of Laplace's equation in cylindrical coordinates is

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \theta^2} + \frac{\partial^2 V}{\partial z^2} = 0. \quad (9)$$

Applying the axisymmetric condition eliminates the angular derivative, resulting in

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{\partial^2 V}{\partial z^2} = 0, \quad (10)$$

which is the governing equation solved numerically in the present work using the Finite Difference Method.

4.4 Boundary Conditions

To obtain a unique solution of Laplace's equation, appropriate boundary conditions must be specified throughout the computational domain. These conditions define either the electric potential or its normal derivative on the domain boundaries.

The computational domain considered in this work consists of three coaxial cylindrical electrodes together with the surrounding vacuum region. Each electrode is assumed to be a perfect conductor; therefore, its surface is maintained at a constant electric potential. The two outer electrodes are grounded, while the central electrode is biased at a potential of V_0 .

Accordingly, the Dirichlet boundary conditions on the electrode surfaces are

$$V = 0, \quad \text{Outer Electrodes} \quad (11)$$

and

$$V = V_0, \quad \text{Central Electrode.} \quad (12)$$

Because the lens possesses axial symmetry, no radial variation of the potential exists along the symmetry axis ($r = 0$). Consequently, the following Neumann boundary condition is imposed:

$$\frac{\partial V}{\partial r} = 0, \quad r = 0. \quad (13)$$

The outer boundary of the computational domain is selected sufficiently far from the lens so that the electric field becomes negligible. Therefore, the electric potential at this boundary is also assumed to be zero,

$$V = 0, \quad \text{Outer Boundary.} \quad (14)$$

These boundary conditions completely define the boundary-value problem and enable the numerical solution of Laplace's equation.

5 Problem Formulation

5.1 Computational Domain

For numerical simulation, the physical space surrounding the electrostatic lens is transformed into a finite computational domain. Although the electric field extends to infinity in reality, an infinite domain cannot be represented directly in numerical calculations. Therefore, the computational boundaries are placed sufficiently far from the lens so that their influence on the solution becomes negligible.

Due to the axial symmetry of the lens, only the half-plane corresponding to $r \geq 0$ is considered. This significantly reduces the computational cost while preserving the physical characteristics of the system.

The computational domain is selected to include the entire lens geometry together with a sufficiently large surrounding vacuum region to accurately represent the electric field distribution.

5.2 Modeling Assumptions

To simplify the numerical model while retaining the essential physical behavior of the electrostatic lens, the following assumptions are adopted:

- The space inside the lens is assumed to be a perfect vacuum.
- Space-charge effects caused by electron-electron interactions are neglected.
- External magnetic fields are ignored.
- Electron motion is treated as non-relativistic.
- The electrodes are modeled as perfect conductors maintained at constant electric potentials.
- The system is assumed to be axisymmetric.
- Quantum mechanical effects and electromagnetic radiation are neglected.

5.3 Simulation Parameters

The numerical simulation requires a set of geometrical, electrical, and numerical parameters to define the physical model and the computational procedure. These parameters describe the lens geometry, the initial conditions of the electron beam, and the numerical settings adopted for solving the governing equations.

The geometrical parameters include the inner radius of the electrodes, the length of each electrode, and the spacing between adjacent electrodes. These quantities determine the overall

geometry of the lens and directly influence the resulting electric field distribution and focusing characteristics.

The electrical parameters consist primarily of the potential applied to the central electrode and the initial kinetic energy of the injected electrons. Variations in these parameters modify the electric field configuration and consequently alter the electron trajectories.

In addition to the physical parameters, several numerical parameters, including the computational domain size, grid resolution, convergence criterion, and maximum number of iterations, are specified for solving Laplace's equation. Appropriate selection of these parameters is essential to ensure the accuracy and stability of the numerical solution.

A summary of the simulation parameters is presented in Table 1.

Table 1: Simulation parameters used in the numerical model.

Parameter	Symbol	Value
Electrode radius	R	5 mm
Electrode length	L	8 mm
Electrode spacing	d	2 mm
Central electrode voltage	V_0	600 V
Initial electron energy	E_k	100 eV
Domain height	$R_{\max}$	15 mm
Domain length	$Z_{\max}$	40 mm
Number of radial grid points	N_r	151
Number of axial grid points	N_z	401
Grid spacing	$\Delta r = \Delta z$	0.1 mm
Relaxation factor	ω	1.92
Convergence tolerance	ε	10^{-3}
Maximum iterations	$N_{\max}$	1000
Number of particles	N_p	11
Time step	Δt	5×10^{-12} s
Maximum simulation time	$t_{\max}$	8×10^{-8} s

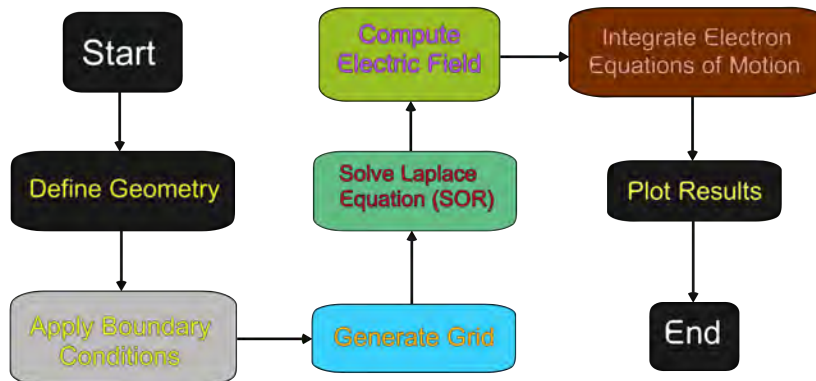


Figure 3: Overall workflow of the numerical simulation procedure.

6 Numerical Implementation

6.1 Grid Generation

After defining the computational domain, it must be discretized into a finite number of grid points to enable the numerical solution of Laplace's equation. In the Finite Difference Method, the derivatives appearing in the governing equations are approximated by algebraic expressions involving neighboring grid points. Consequently, the quality of the computational grid directly affects the accuracy of the numerical solution.

In the present work, a uniform rectangular grid is employed in both the radial and axial directions. The grid spacing in the radial and axial directions is denoted by Δr and Δz , respectively. Owing to the axial symmetry of the system, only the half-plane corresponding to $r \geq 0$ is discretized.

Each grid node represents the electric potential at a specific location inside the computational domain. During the iterative solution process, the potential values at the interior nodes are continuously updated, whereas the boundary nodes remain fixed according to the prescribed boundary conditions.

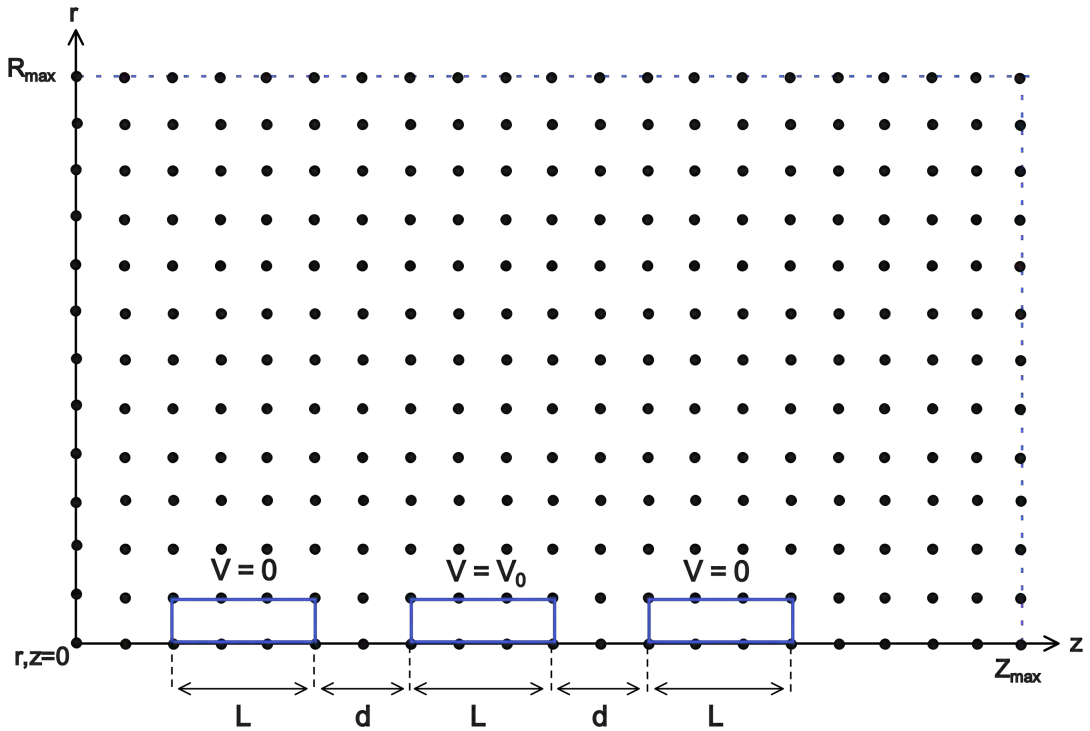


Figure 4: Computational domain and structured finite-difference grid used for solving Laplace's equation. The electrode boundaries are embedded within the computational mesh, while the outer boundary defines the limits of the simulation domain.

Figure 4 presents the computational grid used for discretizing the solution domain.

6.2 Finite Difference Discretization

The objective of the Finite Difference Method is to transform the governing differential equation into a system of algebraic equations that can be solved numerically. This is achieved by replacing the spatial derivatives with finite-difference approximations involving neighboring grid points.

As established in the previous chapter, the governing equation for the electric potential is

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{\partial^2 V}{\partial z^2} = 0. \quad (15)$$

To solve this equation numerically, the computational domain is discretized into a structured grid, and the derivatives are approximated using second-order central finite differences.

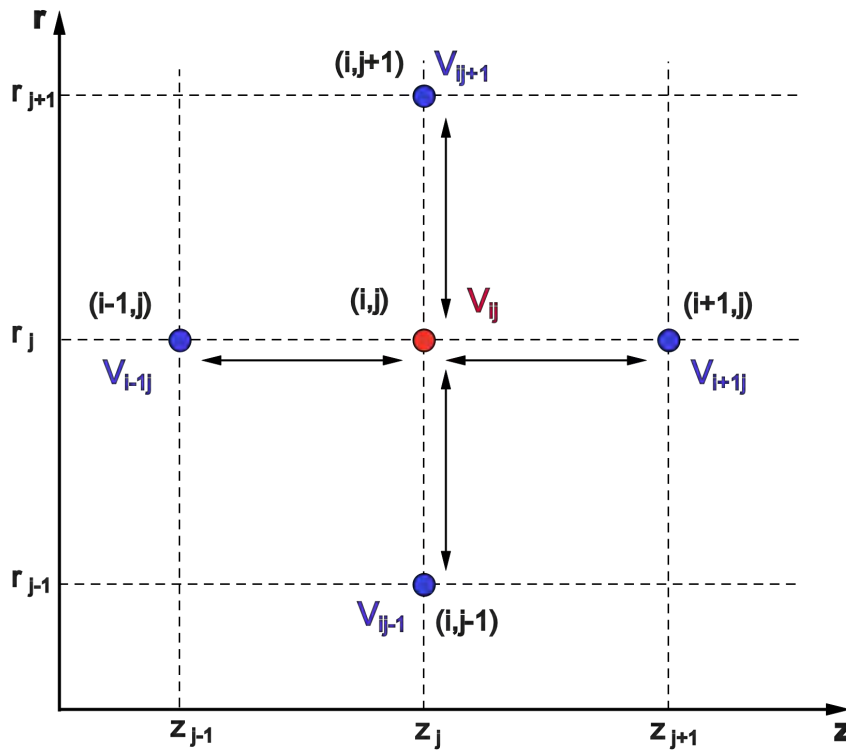


Figure 5: Five-point finite-difference stencil used for evaluating the potential at an interior grid point. The unknown potential at the central node is updated using the values of its neighboring nodes.

As illustrated in Figure 5, the finite-difference approximation employs the four neighboring grid points surrounding the current node. This local stencil forms the basis of the iterative solution procedure used throughout the numerical simulation.

The second derivative in the axial direction is approximated as

$$\frac{\partial^2 V}{\partial z^2} \approx \frac{V_{i,j+1} - 2V_{i,j} + V_{i,j-1}}{\Delta z^2}, \quad (16)$$

whereas the radial operator is approximated by

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) \approx \frac{V_{i+1,j} - 2V_{i,j} + V_{i-1,j}}{\Delta r^2} + \frac{1}{r_i} \frac{V_{i+1,j} - V_{i-1,j}}{2\Delta r}. \quad (17)$$

Substituting these finite-difference approximations into Laplace's equation yields an algebraic equation at every interior grid point. The resulting linear system is solved iteratively to obtain the electric potential distribution throughout the computational domain.

6.3 Update Equation

Following the finite-difference discretization, an algebraic equation is obtained for every interior grid point. Rearranging this equation with respect to the unknown potential at the current node yields the update formula employed during the iterative solution.

For simplicity, a uniform grid is adopted such that

$$\Delta r = \Delta z = h. \quad (18)$$

Under this assumption, the discretized Laplace equation can be written as

$$V_{i,j} = \frac{1}{4} \left[\left(1 + \frac{h}{2r_i} \right) V_{i+1,j} + \left(1 - \frac{h}{2r_i} \right) V_{i-1,j} + V_{i,j+1} + V_{i,j-1} \right]. \quad (19)$$

This expression indicates that the potential at each interior node is determined from the surrounding neighboring nodes. The coefficients associated with the radial neighbors depend on the radial position owing to the cylindrical coordinate system, whereas the axial neighbors retain constant coefficients.

Equation (19) constitutes the fundamental update relation employed throughout the iterative numerical solution.

6.4 Iterative Solution Using SOR

Following the finite-difference discretization, a large system of algebraic equations is obtained for the interior grid points. Directly solving this system is computationally expensive because of the large number of unknowns. Therefore, an iterative method is employed to compute the electric potential distribution.

The Successive Over-Relaxation (SOR) method is one of the most widely used iterative algorithms for solving elliptic partial differential equations such as Laplace's equation. It represents an improved version of the Gauss-Seidel method by introducing a relaxation factor that significantly accelerates convergence.

The relaxation factor used in the present simulation is

$$\omega = 1.8$$

which was selected empirically to accelerate convergence while maintaining numerical stability.

In the SOR method, the potential at each grid point is updated according to

$$V_{i,j}^{(new)} = (1 - \omega) V_{i,j}^{(old)} + \omega V_{i,j}^{(update)}, \quad (20)$$

where $V^{(old)}$ denotes the potential from the previous iteration, $V^{(update)}$ is the value obtained from the finite-difference update equation, and ω is the relaxation factor.

When $\omega = 1$, the SOR algorithm reduces to the standard Gauss–Seidel method. In practice, values satisfying $1 < \omega < 2$ generally improve the convergence rate, although excessively large values may lead to numerical instability.

The iterative process is terminated when the maximum change in the electric potential between two successive iterations satisfies

$$\max \left| V^{(new)} - V^{(old)} \right| < \varepsilon, \quad (21)$$

where ε is the prescribed convergence tolerance.

6.5 Electric Field Calculation

After obtaining the electric potential distribution, the electric field is calculated as the negative gradient of the potential. Since the force acting on an electron depends directly on the electric field, this step provides the essential input for particle trajectory calculations.

According to electrostatic theory,

$$\mathbf{E} = -\nabla V \quad (22)$$

In cylindrical coordinates,

$$E_r = -\frac{\partial V}{\partial r} \quad (23)$$

$$E_z = -\frac{\partial V}{\partial z} \quad (24)$$

The derivatives are evaluated numerically using second-order central differences,

$$E_r(i, j) = -\frac{V_{i+1,j} - V_{i-1,j}}{2\Delta r} \quad (25)$$

$$E_z(i, j) = -\frac{V_{i,j+1} - V_{i,j-1}}{2\Delta z}. \quad (26)$$

6.6 Particle Trajectory Calculation

After computing the electric field distribution, the next step is to determine the trajectories of electrons inside the electrostatic lens. The force acting on each electron is calculated from the electric field, and the equations of motion are integrated numerically.

Since the electric field is available only at the computational grid points, bilinear interpolation is employed to estimate the electric field at the instantaneous position of each electron. This provides a smooth and continuous field evaluation during particle motion.

The force acting on an electron is given by

$$\mathbf{F} = q\mathbf{E}, \quad (27)$$

where q is the electron charge and $\mathbf{E}$ is the electric field.

Applying Newton's second law yields

$$m \frac{d\mathbf{v}}{dt} = q\mathbf{E}, \quad (28)$$

where m denotes the electron mass.

Under the axisymmetric assumption, the equations of motion become

$$\frac{dv_r}{dt} = \frac{q}{m} E_r, \quad (29)$$

$$\frac{dv_z}{dt} = \frac{q}{m} E_z. \quad (30)$$

The particle position is then updated according to

$$r^{n+1} = r^n + v_r \Delta t, \quad (31)$$

$$z^{n+1} = z^n + v_z \Delta t. \quad (32)$$

In the present implementation, the equations of motion are integrated using the explicit Forward Euler method. During each time step, the electric field is first interpolated at the particle position, after which the particle velocity and position are updated sequentially.

The integration continues until the particle exits the computational domain or collides with an electrode surface.

To evaluate the focusing behavior of the electrostatic lens, multiple electrons with different initial radial positions are simulated simultaneously. The resulting trajectories are subsequently used to determine the beam envelope and the focal position of the lens.

6.7 Beam Envelope and Focal Point Determination

Following the particle trajectory calculations, the focusing performance of the electrostatic lens is evaluated by analyzing the evolution of the beam radius along the propagation direction.

At each axial position, the radial positions of all simulated electrons are collected. Instead of using the maximum beam radius, which may be affected by a small number of outlier particles, the beam envelope is defined using the 95th percentile of the particle radii. This provides a more robust estimate of the effective beam size.

Accordingly, the beam envelope is defined as

$$R_b(z) = P_{95}(r(z)), \quad (33)$$

where P_{95} denotes the 95th percentile of the particle radial positions at a given axial location.

The focal point of the electrostatic lens is then identified as the axial position corresponding to the minimum beam radius,

$$z_f = \arg \min R_b(z), \quad (34)$$

and the corresponding focal radius is

$$R_f = R_b(z_f). \quad (35)$$

This approach provides a reliable quantitative measure of the focusing performance while reducing the influence of a small number of abnormal trajectories. The calculated beam envelope and focal position are subsequently used to compare different lens configurations and operating conditions.

7 Software Architecture

The numerical simulation was implemented in Python using a modular programming approach. Instead of placing the entire implementation in a single source file, the program was divided into several independent modules, each responsible for a specific task within the simulation. This organization improves code readability, simplifies debugging, facilitates future modifications, and enables individual components of the program to be tested independently.

The overall workflow begins by loading the simulation parameters and physical constants. The computational domain and lens geometry are then generated, after which the boundary conditions are imposed on the electrode surfaces. Laplace's equation is subsequently solved using the Successive Over-Relaxation (SOR) method to obtain the electric potential distribution. Once the potential converges, the electric field is calculated numerically from the spatial gradient of the potential. Finally, the equations of motion are integrated to determine the trajectories of the electrons, and all simulation results are visualized automatically.

7.1 Program Structure

The software consists of several Python modules, each dedicated to a specific stage of the numerical simulation. Separating the implementation into multiple files increases maintainability and makes the simulation easier to understand and extend.

Table 2 summarizes the primary responsibilities of each module.

Table 2: Main software modules and their functions.

Module	Description
constants.py	Physical constants and simulation parameters
geometry.py	Construction of the lens geometry and electrode masks
solver.py	Numerical solution of Laplace's equation using SOR
field.py	Electric field calculation from the potential distribution
particles.py	Initialization of electron beam parameters
trajectory.py	Numerical integration of electron trajectories
plotting.py	Generation of figures and graphical outputs
main.py	Main program controlling the simulation workflow

7.2 Overall Computational Procedure

The complete numerical simulation follows a sequential workflow in which the output of each computational stage serves as the input for the subsequent stage. This organization ensures that the electric field is obtained from a physically consistent potential distribution before particle motion is evaluated.

The principal computational steps are summarized below:

1. Read all physical constants and simulation parameters.
2. Generate the computational grid and define the lens geometry.
3. Apply Dirichlet and Neumann boundary conditions.
4. Solve Laplace's equation iteratively using the Successive Over-Relaxation method.

5. Monitor convergence until the prescribed tolerance is reached.
6. Compute the electric field from the numerical gradient of the potential.
7. Initialize the electron beam with the desired energy and divergence.
8. Integrate the equations of motion for every particle.
9. Store particle trajectories and beam envelope.
10. Generate all graphical outputs automatically.

The entire procedure is executed automatically by the main program without requiring manual intervention after the simulation parameters have been specified.

7.3 Numerical Techniques

Several numerical techniques are combined in order to simulate the electrostatic lens efficiently and accurately.

The electric potential is obtained by solving Laplace's equation using the Finite Difference Method (FDM). Spatial derivatives are approximated with second-order central finite differences on a structured grid.

The resulting system of algebraic equations is solved iteratively using the Successive Over-Relaxation (SOR) method. Compared with the classical Gauss-Seidel method, SOR significantly accelerates convergence by introducing a relaxation factor greater than unity.

After convergence, the electric field components are computed from the numerical gradient of the electric potential using finite-difference approximations.

Electron trajectories are then obtained by numerically integrating Newton's equations of motion. At each time step, the electric field acting on each particle is interpolated from the surrounding grid nodes, and the corresponding acceleration is evaluated from the Lorentz force equation considering only electrostatic forces.

Numerical computations are performed primarily using NumPy arrays, allowing efficient vectorized operations and reducing computational cost. Graphical visualization of the simulation results is carried out using the Matplotlib library.

7.4 Simulation Workflow

Figure 3 illustrates the complete sequence of operations performed during the numerical simulation.

The simulation begins with the definition of the physical parameters, including the electrode geometry, applied voltages, computational domain, and numerical settings. After initialization, the computational mesh is generated and the boundary conditions are assigned to the corresponding grid nodes.

The iterative SOR solver is then executed until the prescribed convergence criterion is satisfied. The resulting electric potential distribution is subsequently used to calculate the electric field throughout the computational domain.

The electron beam is initialized according to the specified energy, divergence angle, and initial positions. The equations of motion are integrated over time to determine the particle trajectories inside the electrostatic lens.

Finally, all numerical results, including the electric potential, electric field, equipotential contours, field streamlines, convergence history, beam envelope, and particle trajectories, are automatically generated and stored as graphical output files.

7.5 Generated Outputs

After completion of the numerical simulation, the program automatically produces a collection of graphical outputs for analysis and visualization of the electrostatic lens performance.

Table 3 summarizes the generated figures.

Table 3: Automatically generated simulation outputs.

Output File	Description
geometry.png	Geometry of the electrostatic lens
potential.png	Electric potential distribution
electric_field.png	Electric field magnitude
equipotential.png	Equipotential contours
streamlines.png	Electric field streamlines
trajectories.png	Electron trajectories
beam_envelope.png	Beam envelope evolution
convergence.png	Convergence history of the SOR solver

These graphical outputs provide a comprehensive visualization of the electrostatic behavior of the lens together with the focusing characteristics of the simulated electron beam.

8 Results

8.1 Computational Geometry

The first step of the numerical simulation is the generation of the computational geometry of the electrostatic Einzel lens. The model consists of three coaxial cylindrical electrodes surrounded by a vacuum region inside a finite computational domain. The two outer electrodes are maintained at ground potential, whereas the central electrode is biased to produce the electric field required for electron-beam focusing.

Figure 6 illustrates the computational geometry generated by the simulation code. This geometry defines the computational domain and provides the basis for imposing the boundary conditions used throughout the numerical solution.

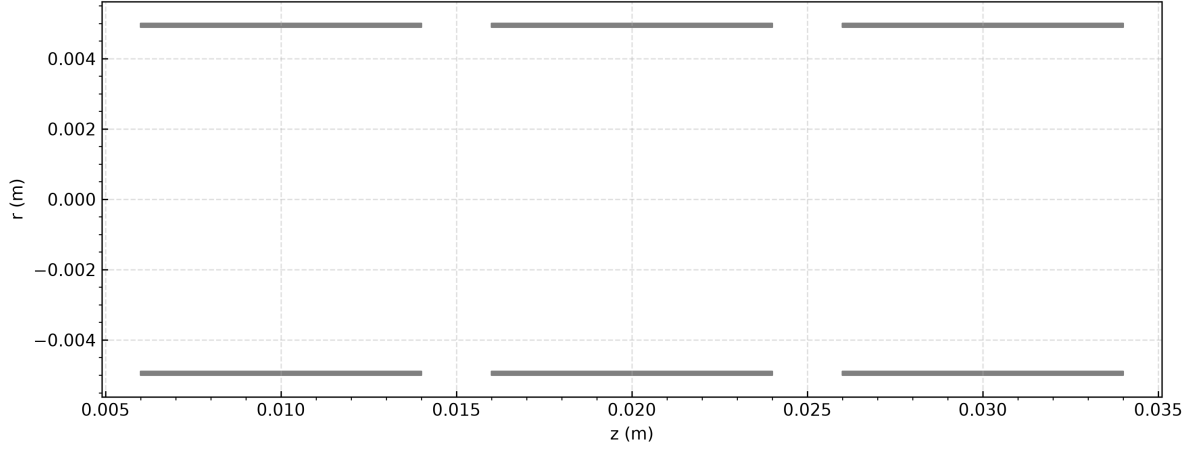


Figure 6: Computational geometry of the electrostatic Einzel lens used in the numerical simulation.

8.2 Electric Potential Distribution

After generating the computational geometry and applying the prescribed boundary conditions, Laplace's equation was solved numerically using the Finite Difference Method together with the Successive Over-Relaxation (SOR) algorithm.

The resulting electric potential distribution is presented in Figure 7. As expected, the highest potential is observed inside the central electrode, while both outer electrodes remain at ground potential. The potential varies smoothly within the vacuum region between the electrodes, producing the electric field responsible for focusing the electron beam.

The obtained potential distribution is symmetric with respect to the optical axis, confirming that the axisymmetric geometry and boundary conditions have been correctly implemented in the numerical model.

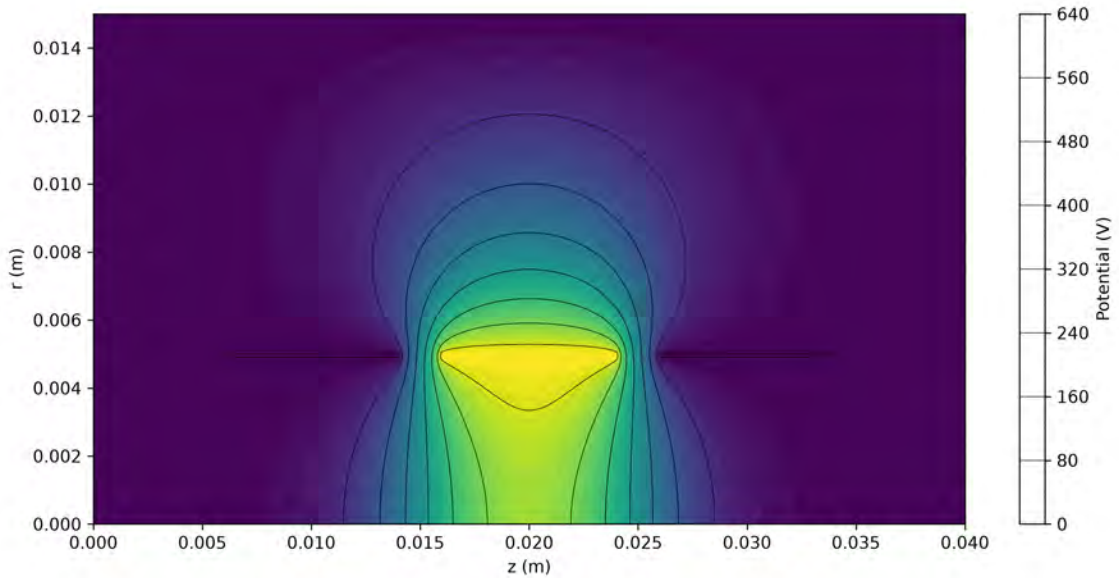


Figure 7: Electric potential distribution obtained from the numerical solution of Laplace's equation.

8.3 Equipotential Lines

A more intuitive representation of the electric potential distribution is obtained by plotting the equipotential lines. These contours connect points having the same electric potential and therefore provide a clear visualization of how the electric potential varies throughout the computational domain.

Figure 8 presents the equipotential contours obtained from the numerical solution. A noticeable concentration of contour lines appears near the central electrode, indicating a larger potential gradient in this region. Since the electric field magnitude is proportional to the spatial gradient of the potential, this behavior corresponds to the region where the electric field is strongest.

The equipotential lines vary smoothly throughout the vacuum region without exhibiting numerical discontinuities or artificial distortions. Furthermore, the contours remain perfectly symmetric about the optical axis, confirming the correctness of both the imposed boundary conditions and the numerical implementation.

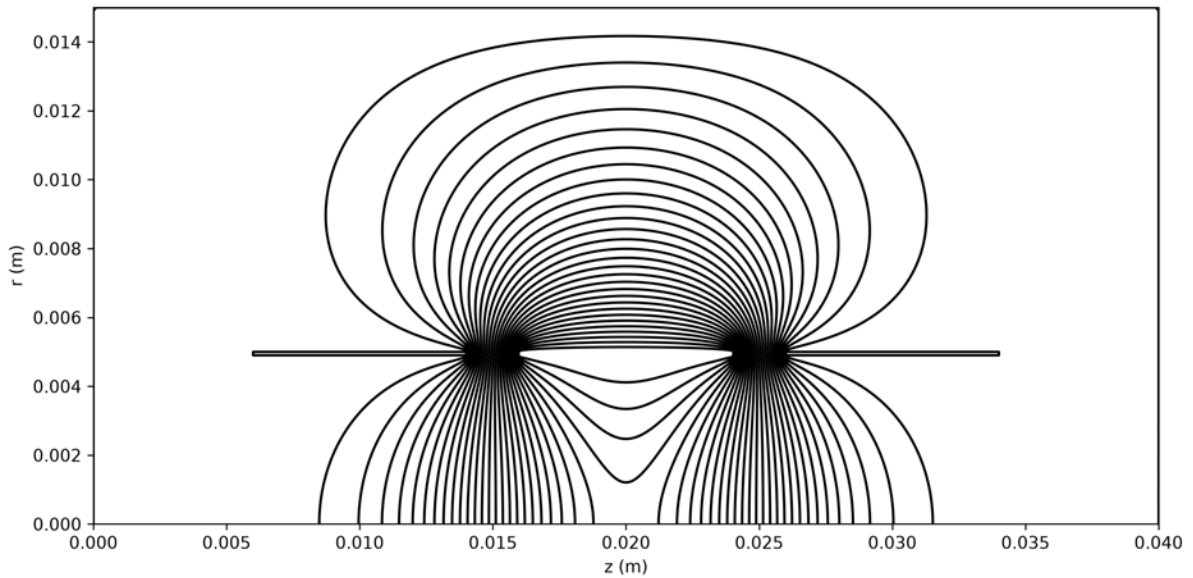


Figure 8: Equipotential contours inside the electrostatic Einzel lens.

8.4 Electric Field Distribution

The electric field was obtained by evaluating the negative gradient of the computed electric potential. Since the electrostatic force acting on each electron is directly proportional to the electric field, this quantity plays the central role in determining the particle trajectories.

Figure 9 shows the magnitude of the electric field throughout the computational domain. As expected, the strongest electric field appears near the edges of the central electrode and within the gaps separating adjacent electrodes. These regions correspond to the largest potential gradients and therefore generate the strongest focusing forces.

Away from the electrodes, the electric field gradually decreases toward zero, indicating that the computational domain is sufficiently large for the imposed outer boundary conditions to have negligible influence on the solution. The field distribution also remains symmetric with respect to the optical axis, confirming the consistency of the numerical implementation.

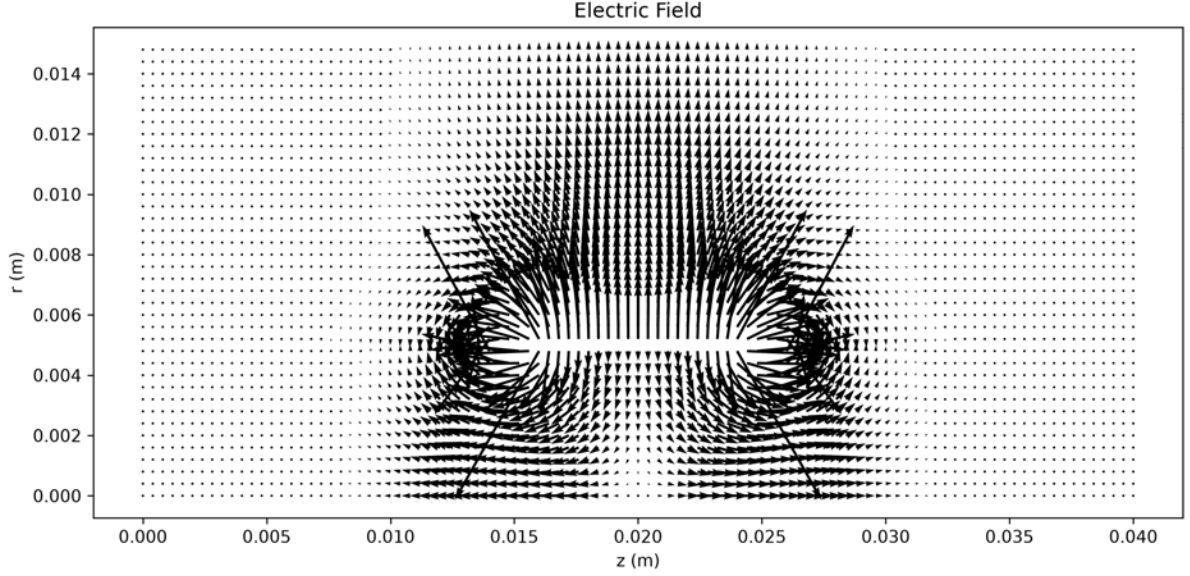


Figure 9: Magnitude of the electric field inside the electrostatic Einzel lens.

8.5 Electric Field Streamlines

While the electric field magnitude provides information about the field strength, the field streamlines illustrate the direction of the electric field throughout the computational domain. These lines are everywhere tangent to the electric field vector and are perpendicular to the equipotential contours.

Figure 10 presents the electric field streamlines generated from the numerical solution. The streamlines become denser near the gaps between adjacent electrodes and around the edges of the central electrode, indicating regions of stronger electric field.

Since electrons carry negative charge, the electrostatic force acting on them is opposite to the direction of the electric field lines. Consequently, the curvature of the field lines near the central electrode produces the focusing action responsible for steering the electron beam toward the optical axis.

The streamlines are smooth and exhibit perfect axial symmetry without any artificial discontinuities, providing additional evidence of the stability and accuracy of the numerical solution.

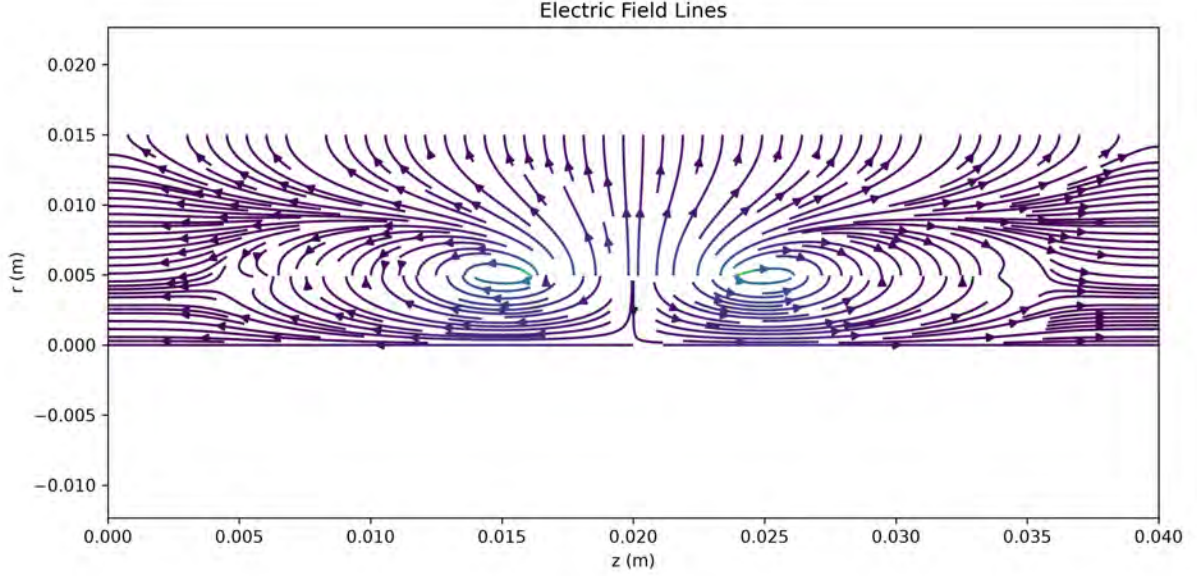


Figure 10: Electric field streamlines inside the electrostatic Einzel lens.

8.6 Electron Trajectories

After computing the electric field, the equations of motion were numerically integrated to determine the trajectories of the electrons inside the electrostatic lens. A set of electrons with identical initial kinetic energy but different initial radial positions was injected into the computational domain.

Figure 11 presents the calculated electron trajectories. Electrons initially located farther from the optical axis experience stronger radial deflection while passing through the electric field. Nevertheless, all trajectories gradually bend toward the optical axis and converge to a common focal region after passing through the lens, demonstrating the focusing capability of the electrostatic Einzel lens.

The particle trajectories remain smooth and continuous throughout the simulation without exhibiting numerical oscillations or discontinuities. This behavior indicates that both the computed electric field and the numerical time integration scheme provide stable and physically consistent particle motion.

Although the trajectories do not intersect at exactly one point, the small spread around the focal region suggests good focusing performance and relatively low optical aberration for the selected operating conditions.

The minimum beam radius occurs at approximately $z = 1.70 \times 10^{-2}$ m, which corresponds to the focal position of the electrostatic lens.

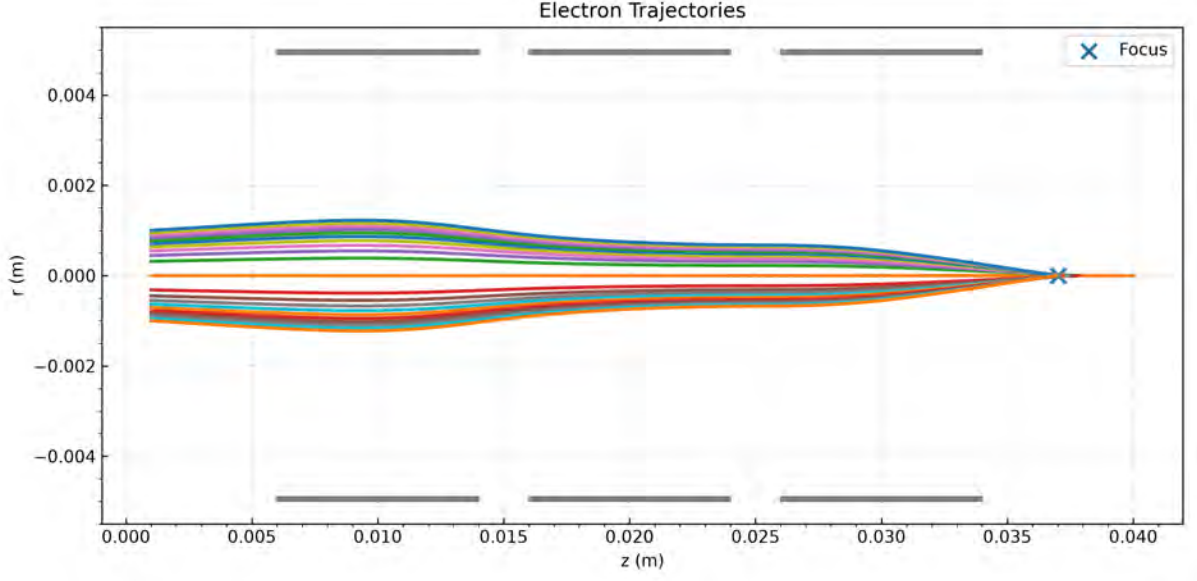


Figure 11: Simulated electron trajectories inside the electrostatic Einzel lens.

8.7 Beam Envelope

To quantitatively evaluate the focusing performance of the electrostatic lens, the beam radius was calculated as a function of the axial position. At each axial location, the beam radius was obtained from the radial distribution of all simulated electrons.

Figure 12 illustrates the evolution of the beam envelope throughout the computational domain. Initially, the beam radius is approximately equal to the injected beam radius. As the electrons enter the electric field generated by the lens, the radial focusing force gradually decreases the beam radius.

The beam reaches its minimum radius at the focal region, indicating the point of strongest focusing. Beyond this location, the beam radius increases again as the electrons diverge after crossing the focal point. This behavior is characteristic of an electrostatic focusing system and confirms the expected operation of the Einzel lens.

Furthermore, the beam envelope varies smoothly along the propagation direction, indicating stable numerical integration and a physically consistent electric-field solution.

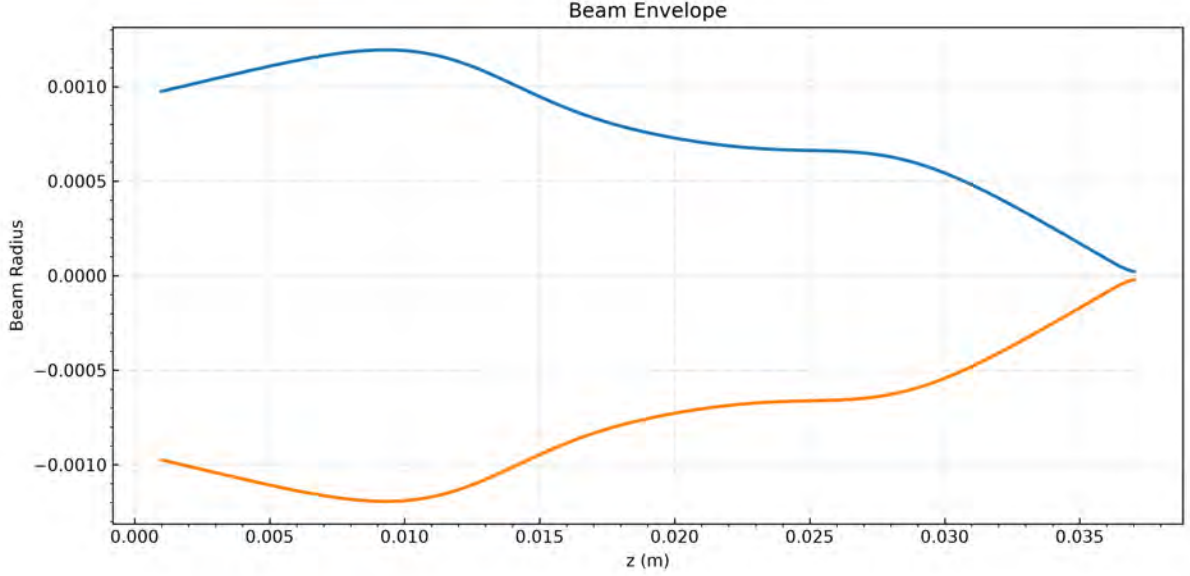


Figure 12: Beam envelope showing the evolution of the electron beam radius along the optical axis.

8.8 Overall Performance of the Electrostatic Lens

Overall, the numerical results demonstrate that the developed model successfully reproduces the expected behavior of an electrostatic Einzel lens. The calculated electric potential satisfies the imposed boundary conditions, while the resulting electric field exhibits the characteristic focusing pattern required for charged-particle manipulation.

The simulated electron trajectories show that the electric field generated by the central electrode gradually deflects the electrons toward the symmetry axis, leading to beam convergence and the formation of a focal region downstream of the lens. The beam-envelope analysis further confirms this behavior by illustrating a continuous reduction of the beam radius followed by beam divergence after the focal point.

The electric-field lines and equipotential contours also indicate that the modified electrode geometry generates a smooth and nearly symmetric electric-field distribution. As expected, the strongest field gradients occur near the gaps between adjacent electrodes, where the focusing action is primarily produced.

Overall, the combination of the Finite Difference Method, the Successive Over-Relaxation algorithm, and numerical particle tracing provides an effective computational framework for the analysis and design of electrostatic focusing systems. The developed simulation can also serve as a basis for future investigations involving different electrode geometries, operating voltages, and beam energies.

9 Parametric Study

9.1 Effect of Central Electrode Voltage

The voltage applied to the central electrode is the primary parameter governing the focusing strength of an electrostatic Einzel lens. By increasing the applied voltage, the electric potential distribution inside the lens changes, producing stronger electric field gradients that modify the

electron trajectories. To investigate this effect, the central electrode voltage was varied while all remaining geometrical and numerical parameters were kept constant.

The investigated voltages were

$$V_0 = 300, 450, 600, 750, 900, 1200 \text{ V.}$$

The corresponding numerical results are presented in the following figures.

9.2 Effect of Central Electrode Voltage

The focusing characteristics of the electrostatic lens strongly depend on the voltage applied to the central electrode. To investigate this effect, simulations were performed for six different voltages while keeping all other parameters constant.

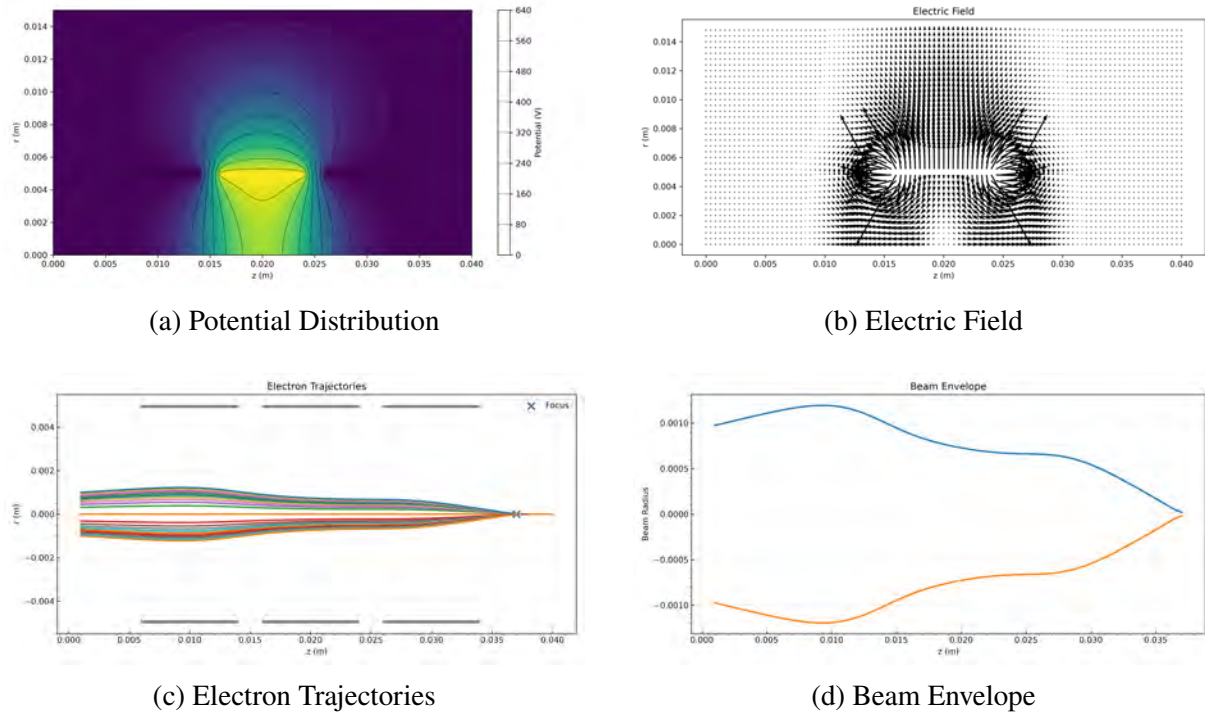
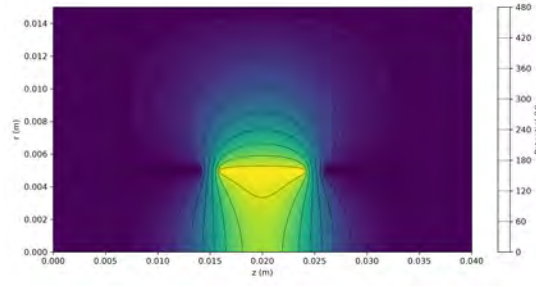
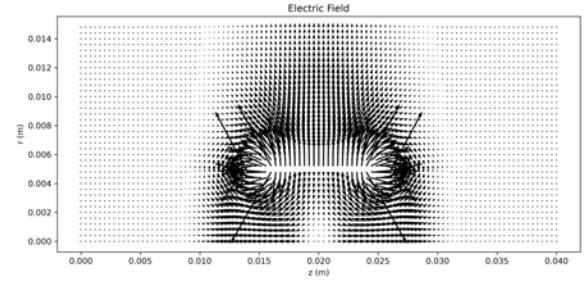


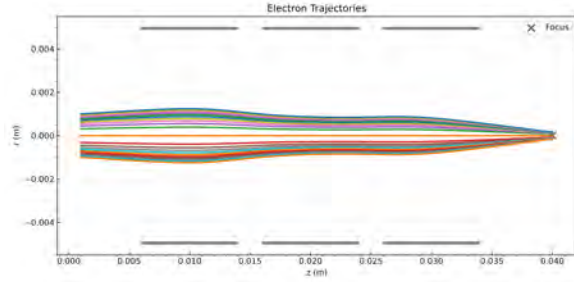
Figure 13: Simulation results for a central electrode voltage of $V_0 = 300 \text{ V}$.



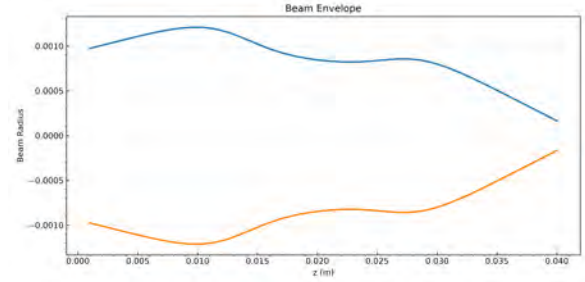
(a) Potential Distribution



(b) Electric Field

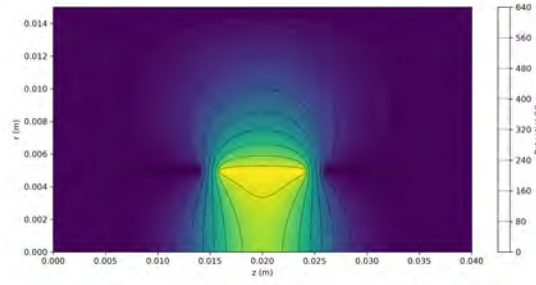


(c) Electron Trajectories

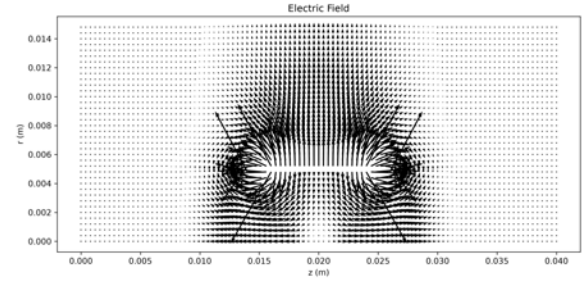


(d) Beam Envelope

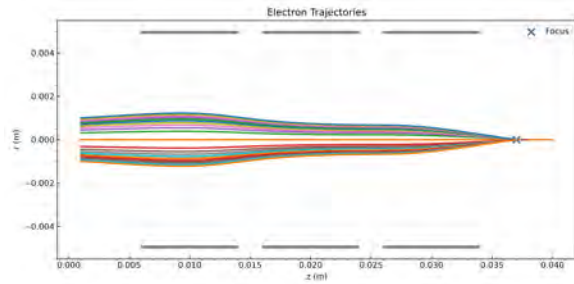
Figure 14: Simulation results for a central electrode voltage of $V_0 = 450$ V.



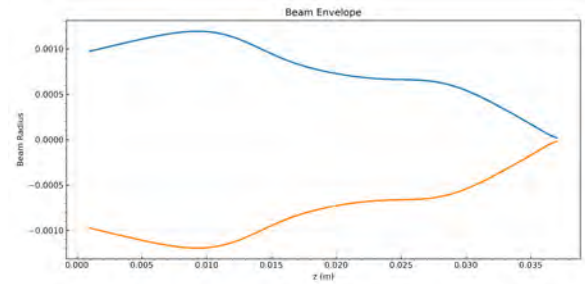
(a) Potential Distribution



(b) Electric Field

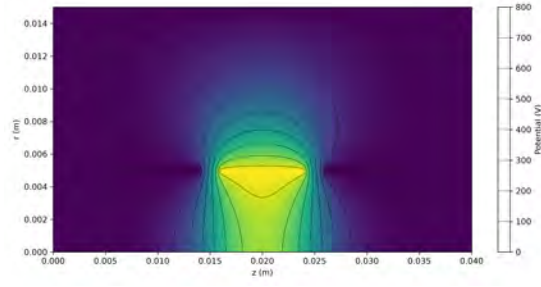


(c) Electron Trajectories

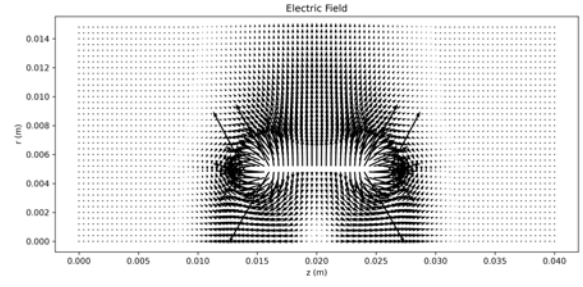


(d) Beam Envelope

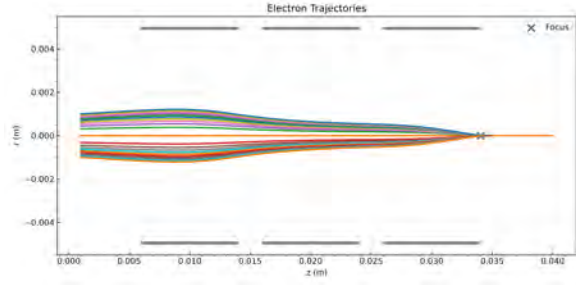
Figure 15: Simulation results for a central electrode voltage of $V_0 = 600$ V.



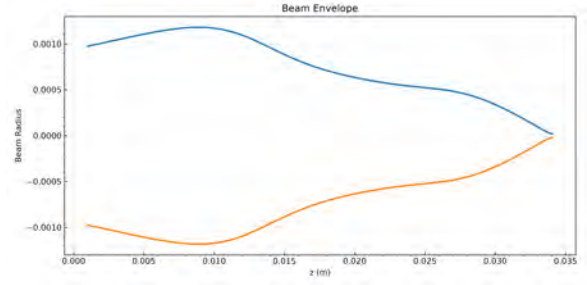
(a) Potential Distribution



(b) Electric Field

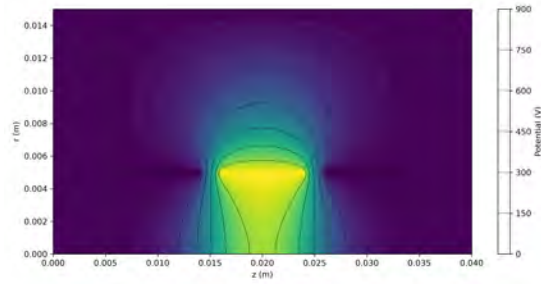


(c) Electron Trajectories

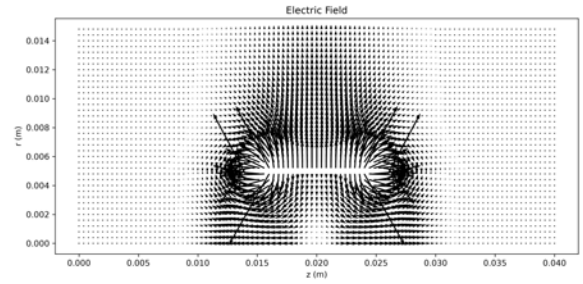


(d) Beam Envelope

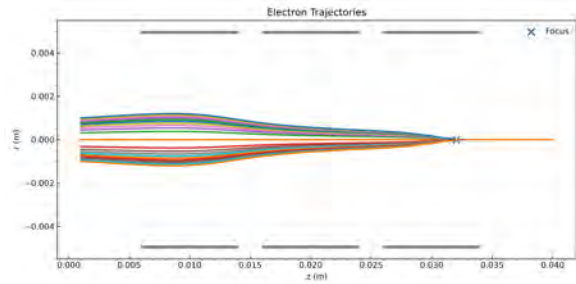
Figure 16: Simulation results for a central electrode voltage of $V_0 = 750$ V.



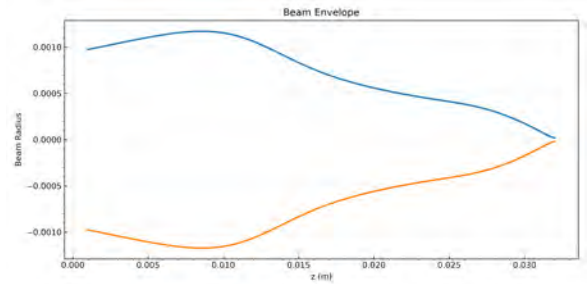
(a) Potential Distribution



(b) Electric Field



(c) Electron Trajectories



(d) Beam Envelope

Figure 17: Simulation results for a central electrode voltage of $V_0 = 900$ V.

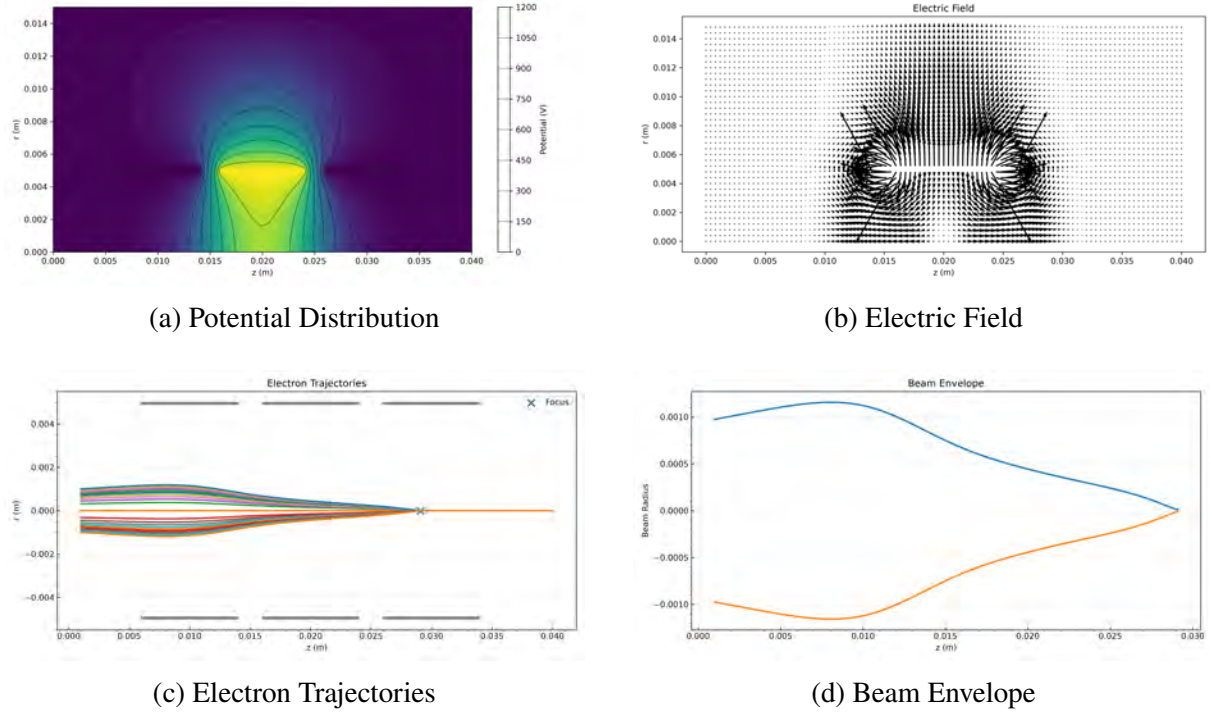


Figure 18: Simulation results for a central electrode voltage of $V_0 = 1200$ V.

Table 4: Comparison of the focusing characteristics for different central electrode voltages.

V_0 (V)	f (mm)	$r_{\min}$ (mm)
300	0.037	≈ 0
450	Outside computational domain	0.0002
600	0.036	≈ 0
750	0.034	≈ 0
900	0.032	≈ 0
1200	0.029	0

The central electrode voltage has the strongest influence on the focusing capability of the electrostatic lens. As the voltage applied to the central electrode increases, the potential difference between the middle electrode and the grounded outer electrodes becomes larger. Consequently, a stronger electric field is established near the electrode gaps, producing a greater radial force on the electrons.

The simulation results clearly show that increasing the central electrode voltage continuously decreases the focal length from approximately 37 mm at 300 V to about 29 mm at 1200 V. This behavior indicates that the lens becomes progressively stronger, causing the electron beam to converge more rapidly.

In addition to reducing the focal length, higher voltages also improve the beam compression. Except for the lowest voltage case, the minimum beam radius is essentially zero, indicating that the beam is nearly focused to a single point. At 300 V the focusing force is relatively weak and the beam converges farther from the lens, whereas at higher voltages the stronger electric field bends the trajectories more efficiently.

The electric potential distributions also reveal a larger potential gradient around the central electrode as the voltage increases. This leads directly to higher electric field magnitudes, which

are responsible for the enhanced focusing action observed in the trajectory and beam-envelope plots.

Overall, the simulations demonstrate that increasing the central electrode voltage strengthens the electrostatic lens, shortens the focal length, and improves beam focusing performance. However, excessively high voltages may become impractical in real systems because of electrical breakdown limitations and increased field emission from electrode surfaces.

9.3 Effect of Electrode Radius

The influence of the electrode inner radius on the electrostatic lens performance was investigated by varying the electrode radius while keeping all other parameters constant. The simulations were carried out for electrode radii of 3, 4, 5, 6, and 7 mm.

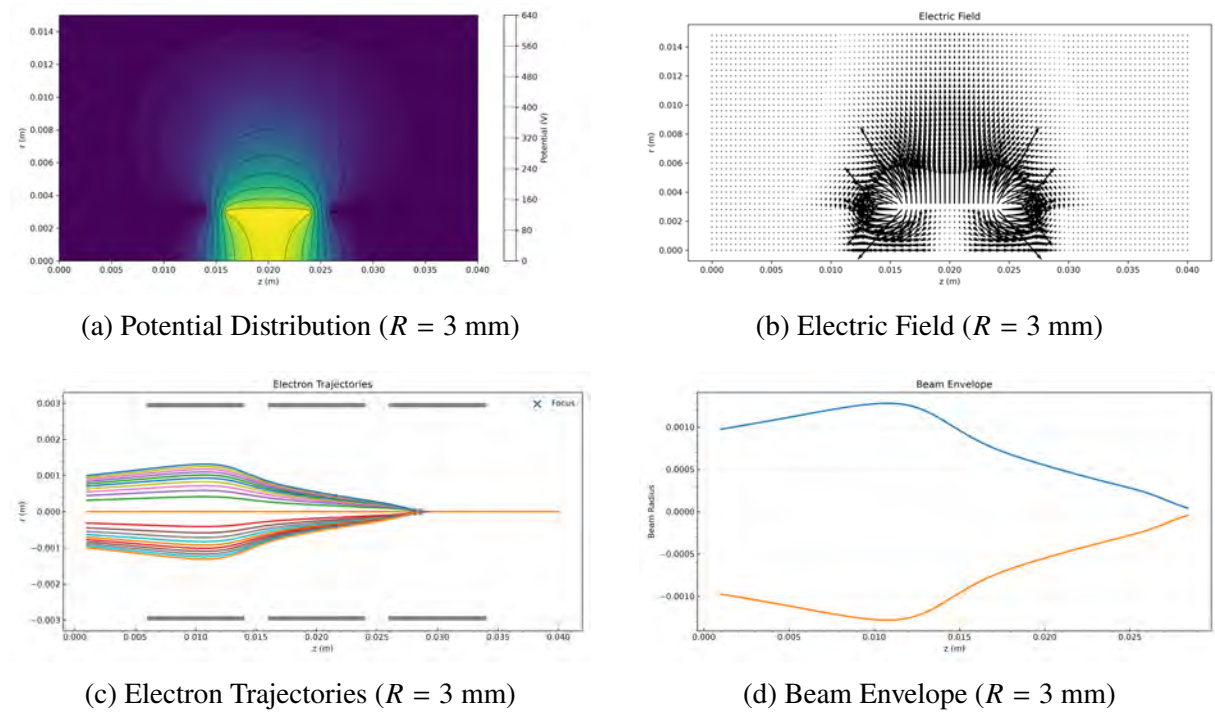
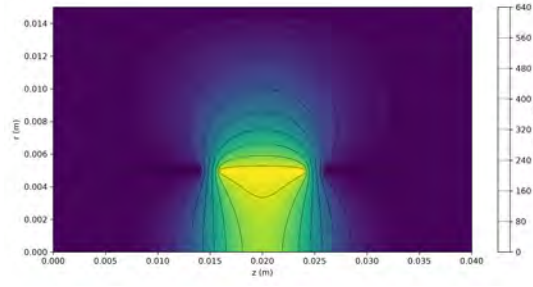
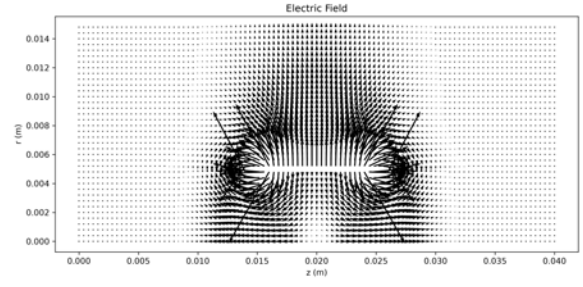


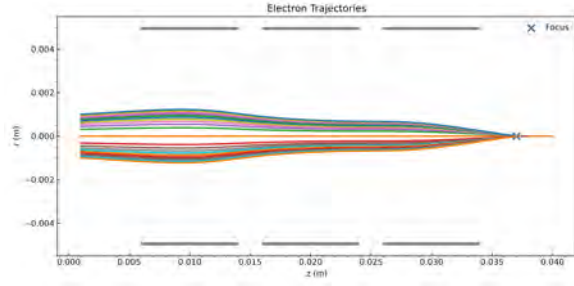
Figure 19: Simulation results for an electrode radius of 3 mm.



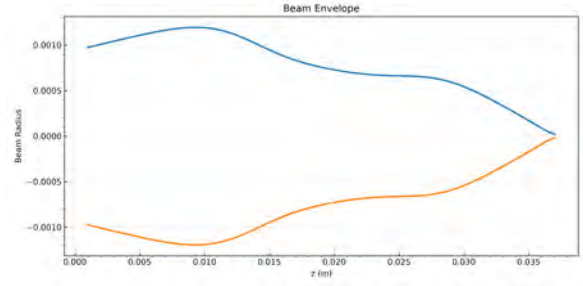
(a) Potential Distribution ($R = 5$ mm)



(b) Electric Field ($R = 5$ mm)

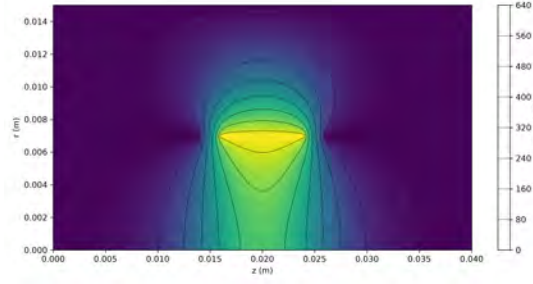


(c) Electron Trajectories ($R = 5$ mm)

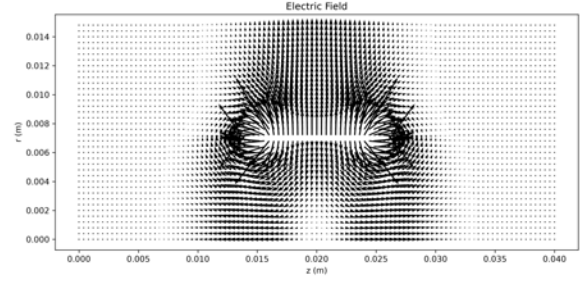


(d) Beam Envelope ($R = 5$ mm)

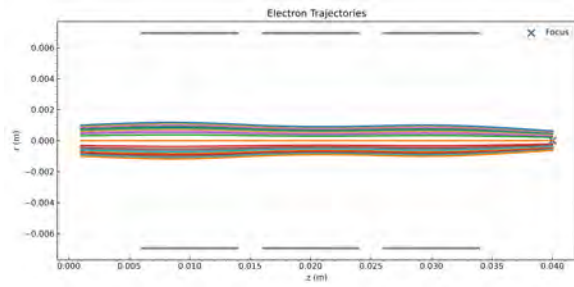
Figure 20: Simulation results for an electrode radius of 5 mm.



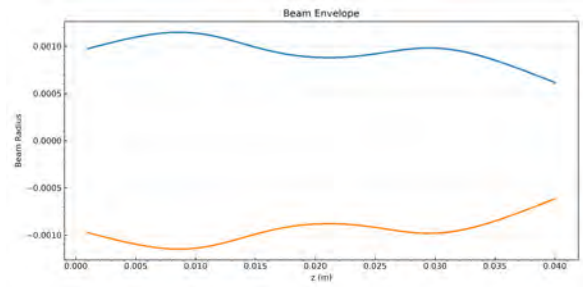
(a) Potential Distribution ($R = 7$ mm)



(b) Electric Field ($R = 7$ mm)



(c) Electron Trajectories ($R = 7$ mm)



(d) Beam Envelope ($R = 7$ mm)

Figure 21: Simulation results for an electrode radius of 7 mm.

Table 5: Comparison of the focusing characteristics for different electrode radii.

R (mm)	f (mm)	$r_{\min}$ (mm)
3	0.0285	0.0001
4	0.033	≈ 0
5	0.0365	≈ 0
6	Outside computational domain	0.0002
7	Outside computational domain(more than before)	0.0006

The electrode radius significantly influences the electric field distribution inside the electrostatic lens. When the radius is reduced, the electrodes are positioned closer to the beam axis, causing the electric field to become more concentrated around the electron trajectories. As a result, the focusing force acting on the electrons increases.

The numerical results show that decreasing the electrode radius from 7 mm to 3 mm substantially shortens the focal length. For the smallest radius, the beam reaches its focal point at approximately 28.5 mm, whereas for larger radii the focal point moves progressively farther away and eventually lies outside the computational domain.

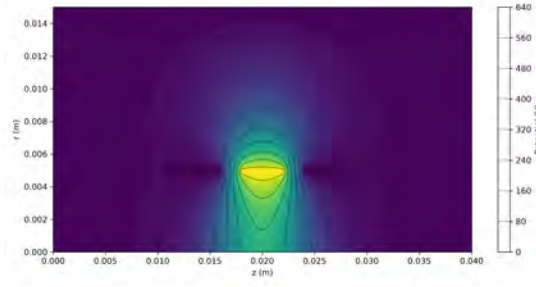
The beam envelope confirms this trend. Small electrode radii produce a very narrow beam waist, while larger radii yield noticeably larger minimum beam radii, indicating weaker focusing. In particular, for a radius of 7 mm the beam is only weakly focused because the electric field is distributed over a larger volume, reducing its radial component near the beam axis.

The electric potential and electric field distributions further support this observation. Smaller electrode radii create steeper potential gradients near the axis, leading to stronger electric fields that efficiently redirect the electron trajectories.

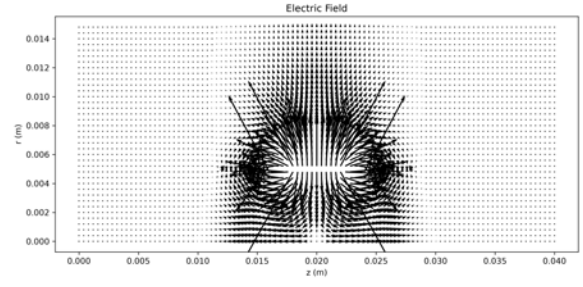
These results indicate that reducing the electrode radius increases the focusing strength of the lens. Nevertheless, extremely small radii may not be desirable in practice because they reduce the available beam aperture and increase the likelihood of beam interception by the electrode surfaces.

9.4 Effect of Electrode Length

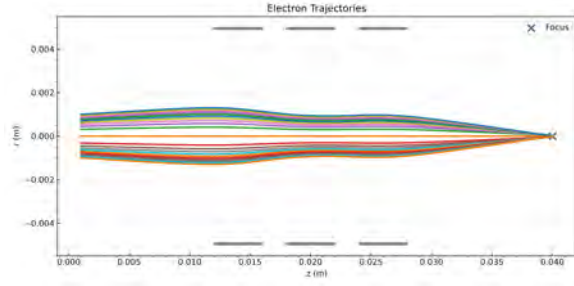
The influence of the electrode length on the electrostatic lens performance was investigated by varying the electrode length while keeping all other parameters constant. The simulations were carried out for electrode lengths of 4, 6, 8, 10, and 12 mm.



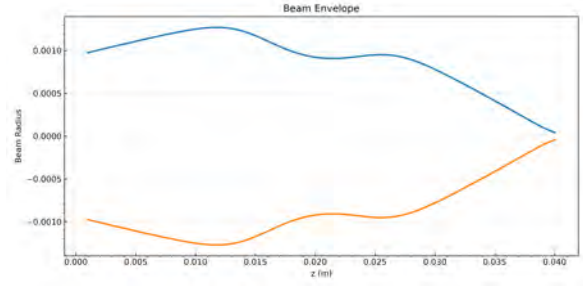
(a) Potential Distribution ($L = 4$ mm)



(b) Electric Field ($L = 4$ mm)

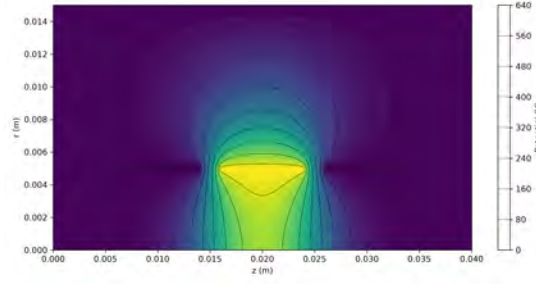


(c) Electron Trajectories ($L = 4$ mm)

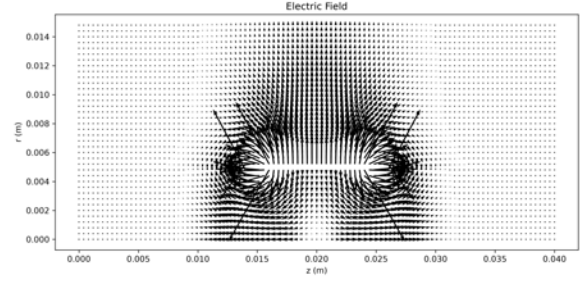


(d) Beam Envelope ($L = 4$ mm)

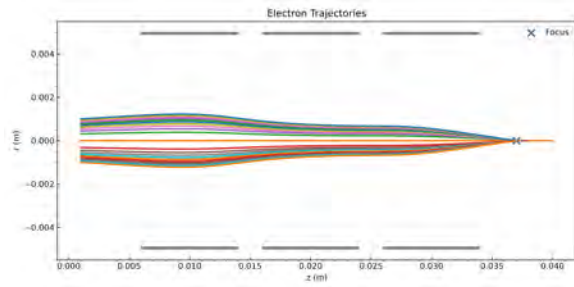
Figure 22: Simulation results for an electrode length of 4 mm.



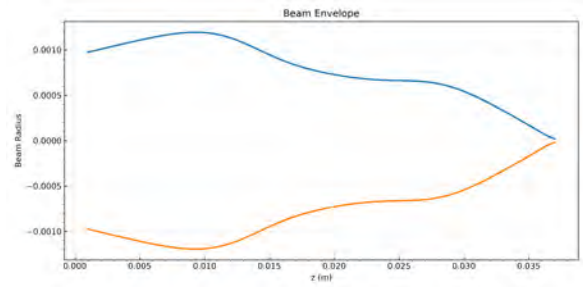
(a) Potential Distribution ($L = 8$ mm)



(b) Electric Field ($L = 8$ mm)



(c) Electron Trajectories ($L = 8$ mm)



(d) Beam Envelope ($L = 8$ mm)

Figure 23: Simulation results for an electrode length of 8 mm.

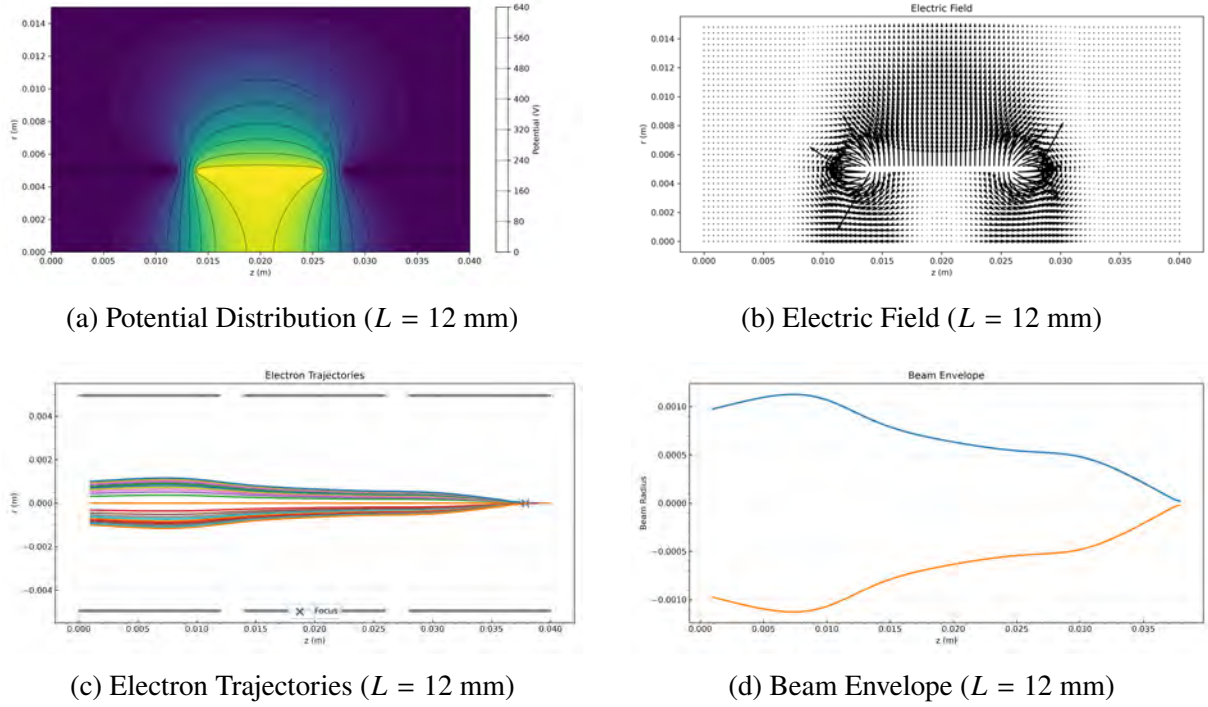


Figure 24: Simulation results for an electrode length of 12 mm.

Table 6: Comparison of the focusing characteristics for different electrode lengths.

L (mm)	f (mm)	$r_{\min}$ (mm)
4	0.040	0.0001
6	0.0375	≈ 0
8	0.037	≈ 0
10	0.037	≈ 0
12	0.038	≈ 0

The length of the central electrode determines the axial distance over which the electrons experience the focusing electric field. Increasing the electrode length extends the interaction region between the beam and the electric field.

The simulations show that changing the electrode length has a relatively small influence on the focusing characteristics compared with the other geometrical parameters. The focal length varies only slightly between approximately 37 mm and 40 mm over the investigated range.

Similarly, the minimum beam radius remains essentially zero for nearly all electrode lengths, indicating that the beam focusing quality is almost unchanged. Only the shortest electrode exhibits a slightly larger minimum beam radius, suggesting a modest reduction in focusing efficiency.

The electric field distributions reveal that increasing the electrode length mainly extends the longitudinal region occupied by the electric field rather than significantly increasing its magnitude. Consequently, the electrons experience the focusing force over a longer distance, but the overall focusing strength remains nearly constant.

Therefore, the electrode length primarily influences the spatial extent of the focusing region rather than the focusing strength itself. The results suggest that moderate variations in electrode length have only a limited effect on lens performance.

9.5 Effect of Gap

The influence of the gap between adjacent electrodes on the electrostatic lens performance was investigated by varying the electrode spacing while keeping all other parameters constant. The simulations were carried out for gap distances of 1, 2, 3, 4, and 5 mm.

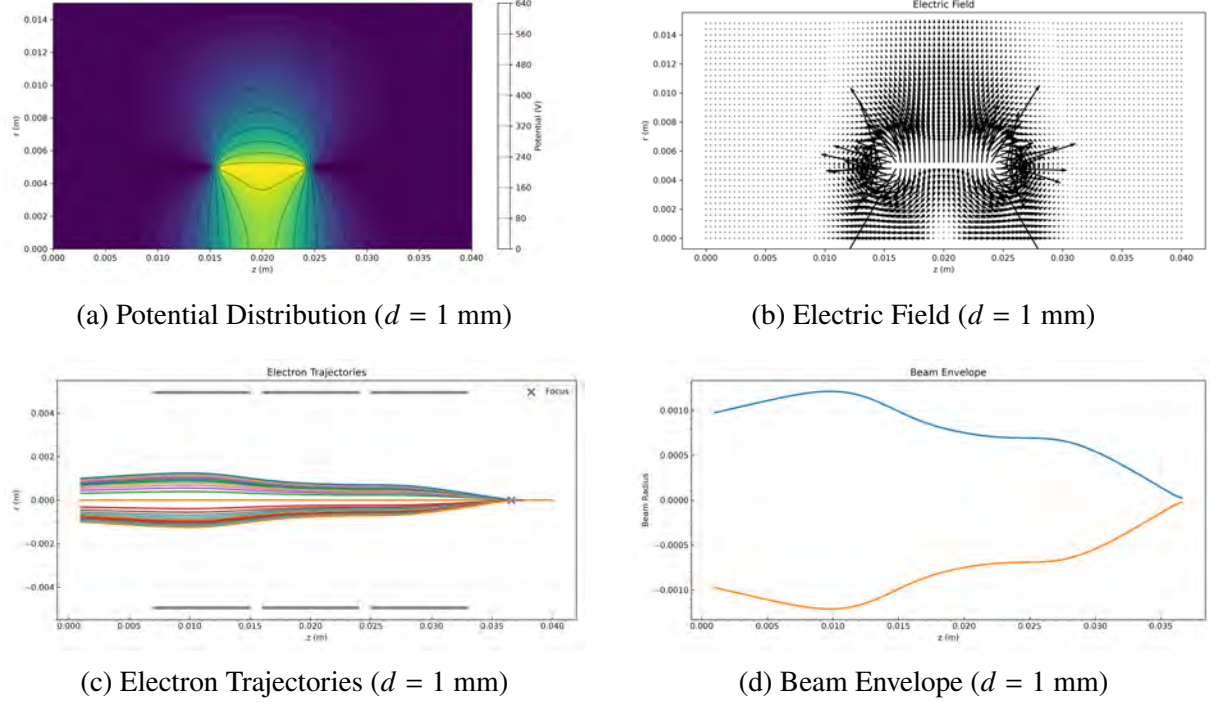


Figure 25: Simulation results for an electrode gap of 1 mm.

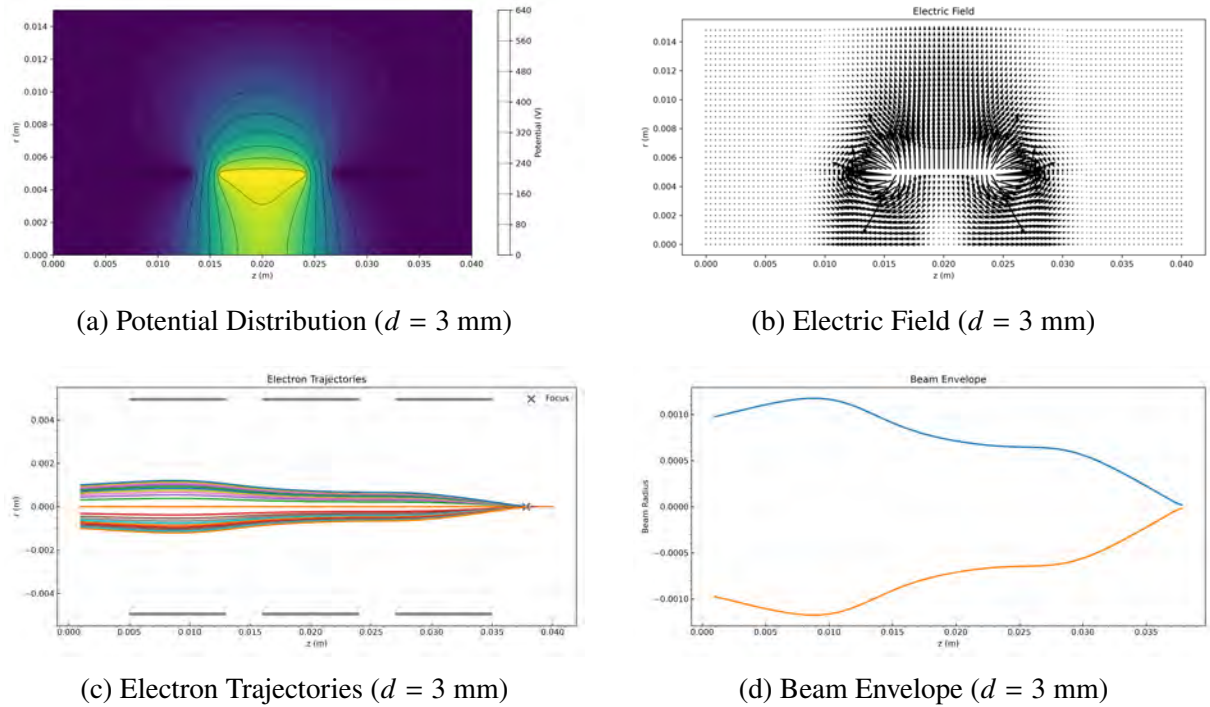


Figure 26: Simulation results for an electrode gap of 3 mm.

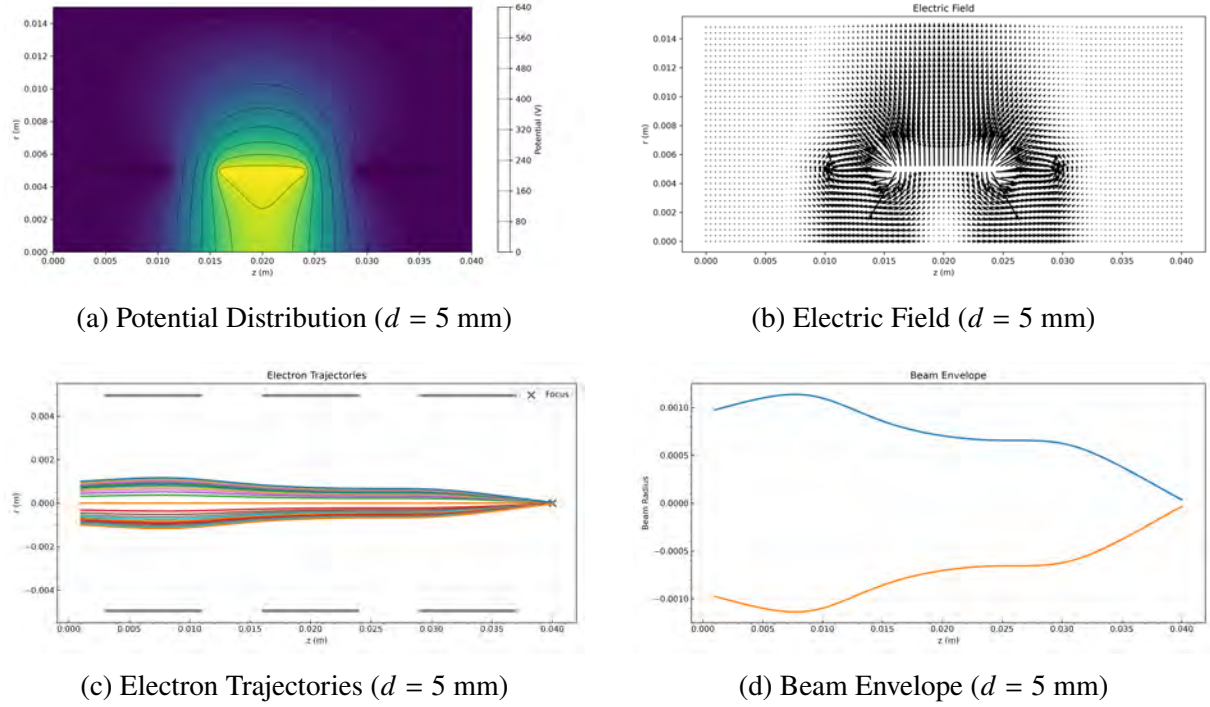


Figure 27: Simulation results for an electrode gap of 5 mm.

Table 7: Comparison of the focusing characteristics for different electrode gaps.

d (mm)	f (mm)	$r_{\min}$ (mm)
1	0.0365	0
2	0.037	0
3	0.038	0
4	0.039	0
5	0.040	0.00008

The spacing between adjacent electrodes controls the transition of the electric potential between neighboring electrodes. A smaller gap produces steeper potential gradients, whereas a larger gap spreads the electric field over a longer distance.

The numerical simulations indicate that increasing the electrode gap gradually increases the focal length from approximately 36.5 mm to about 40 mm. This trend implies that larger gaps reduce the focusing strength of the electrostatic lens.

For small gap values, the electric field is highly concentrated around the electrode edges, generating stronger radial forces that bend the electron trajectories more efficiently. As the gap increases, the electric field becomes more diffuse, resulting in weaker focusing and a more distant focal point.

The beam envelope demonstrates that the beam remains well focused for gap values up to approximately 4 mm. However, at the largest investigated gap, the minimum beam radius begins to increase slightly, indicating a gradual deterioration in focusing quality.

These observations suggest that reducing the electrode spacing enhances the focusing capability by producing stronger electric field gradients. Nevertheless, excessively small gaps may introduce practical manufacturing difficulties and increase the risk of electrical breakdown between electrodes.

9.6 Effect of Initial Electron Energy

The influence of the initial kinetic energy of the electron beam on the focusing performance of the electrostatic lens was investigated by varying the injection energy while keeping all geometrical parameters and electrode voltages constant. The simulations were performed for electron energies of 20, 100, and 800 eV.

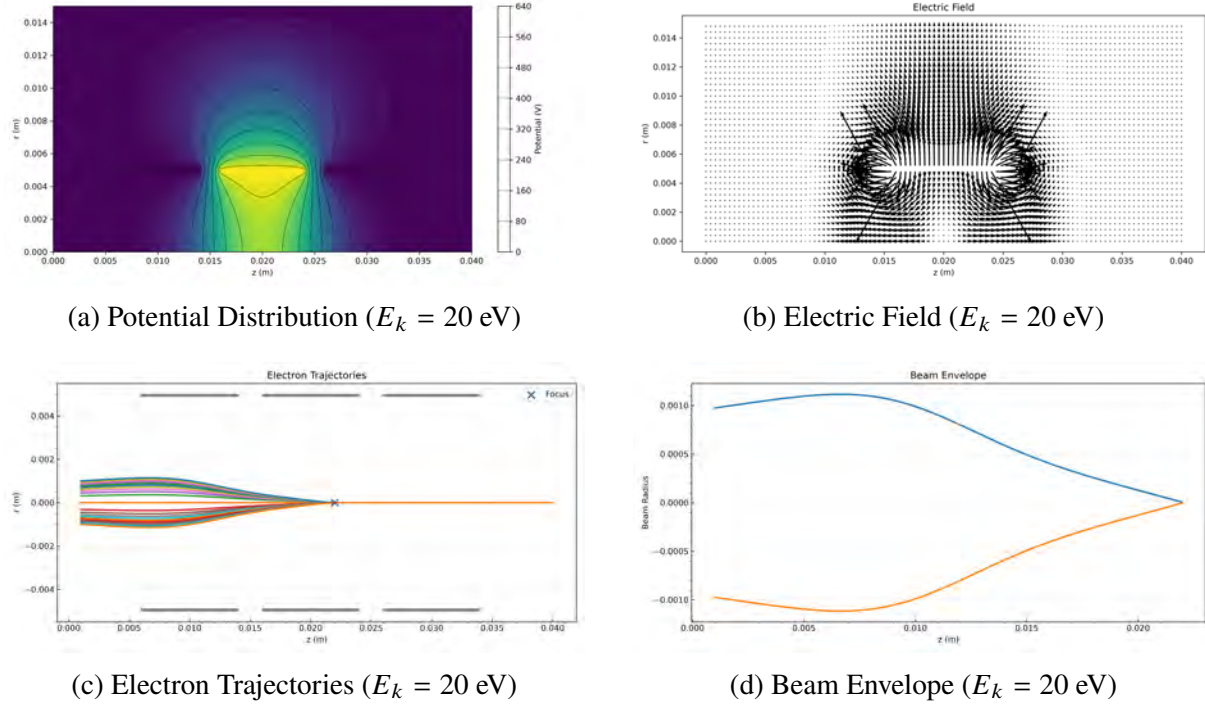


Figure 28: Simulation results for an initial electron energy of 20 eV.

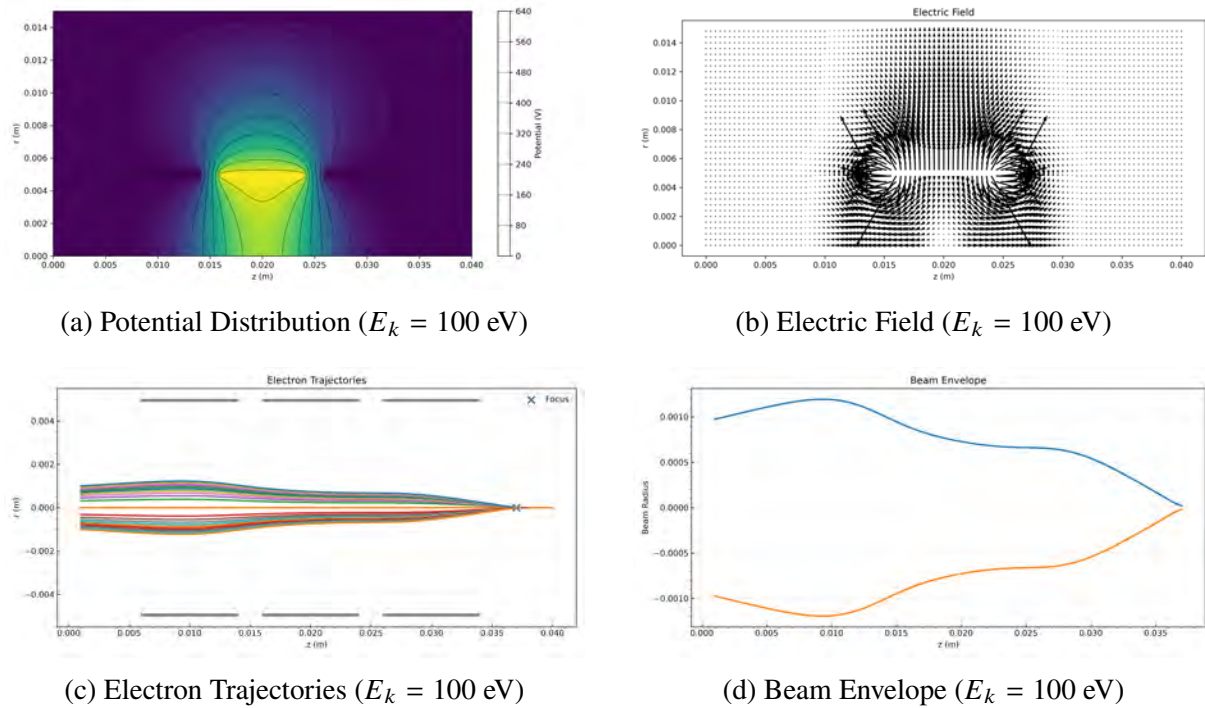


Figure 29: Simulation results for an initial electron energy of 100 eV.

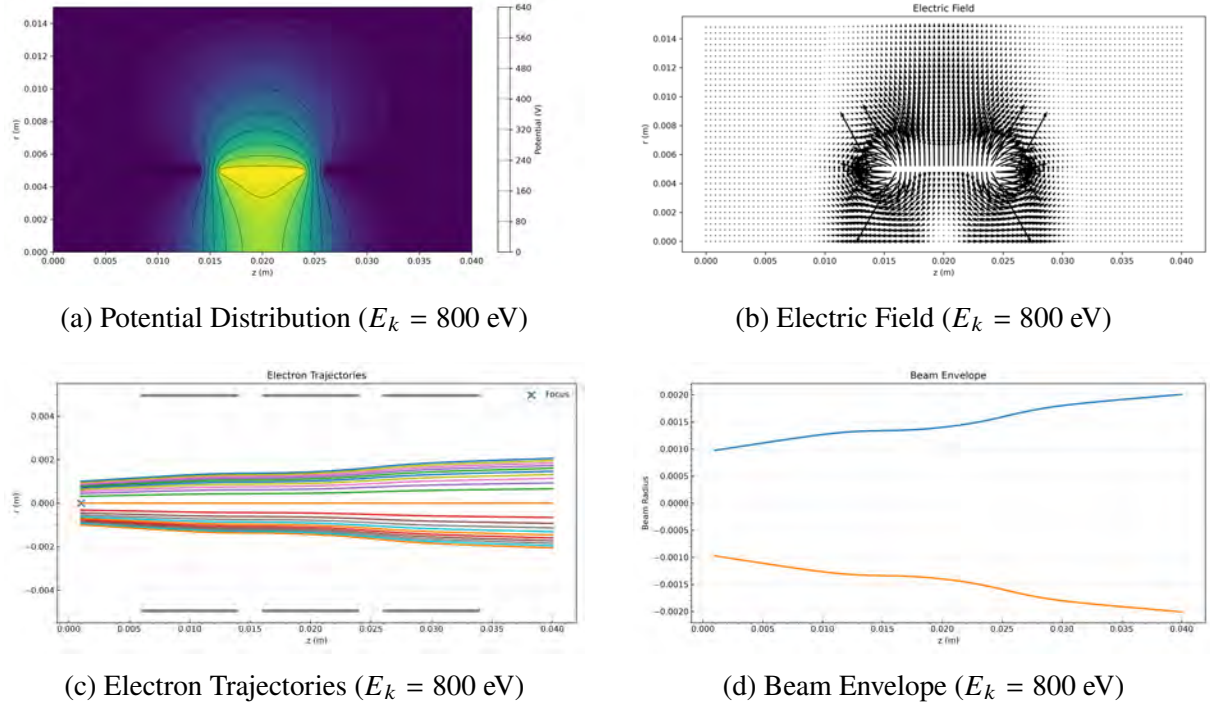


Figure 30: Simulation results for an initial electron energy of 800 eV.

Table 8: Comparison of the focusing characteristics for different initial electron energies.

E_k (eV)	f (mm)	$r_{\min}$ (mm)
20	0.022	0
50	0.029	0
100	0.037	≈ 0
200	Outside computational domain	0.00075
400	Divergent	0.0095
800	Divergent	0.0095

The initial kinetic energy of the electrons has a profound influence on the focusing performance of the electrostatic lens because it determines how strongly the electron trajectories are affected by the electric field.

For low electron energies (20–100 eV), the electrons travel relatively slowly and therefore remain inside the electric field region for a longer period. Consequently, the electric field is able to produce significant radial deflection, resulting in strong beam focusing and short focal lengths.

As the initial energy increases, the electron velocity also increases. Since the electrons spend less time inside the focusing field, the electric force has less opportunity to modify their trajectories. As a result, the focal length increases progressively and the focusing performance deteriorates.

At an initial energy of approximately 200 eV, the focal point already moves beyond the computational domain and the minimum beam radius increases noticeably. This behavior indicates that the electrostatic lens is approaching its focusing limit.

For even higher energies (400 eV and 800 eV), the electric field is no longer sufficiently strong to overcome the electrons' large forward momentum. Consequently, the trajectories never inter-

sect and the beam remains divergent throughout the computational domain. The beam envelope confirms this behavior by showing a continuously expanding beam with no observable waist.

The potential and electric field distributions remain unchanged because the electrode voltages are identical in all simulations. Therefore, the observed changes arise entirely from the increasing electron momentum rather than any modification of the electric field itself.

Overall, these results demonstrate that an electrostatic lens has an effective operating energy range. When the electron energy becomes too large relative to the applied electrode voltage, the focusing force becomes insufficient and the lens loses its ability to focus the beam.

10 Discussion

10.1 Interpretation of the Results

The numerical results presented in the previous chapter indicate that the proposed computational model successfully reproduces the expected physical behavior of an electrostatic Einzel lens. The electric potential varies smoothly throughout the computational domain, while the equipotential contours become increasingly dense near the electrode gaps, indicating regions of large potential gradients.

Since the electric field is obtained from the negative gradient of the electric potential, these regions naturally correspond to the strongest electric fields responsible for focusing the electron beam. The simulated particle trajectories confirm this behavior, showing that electrons are gradually deflected toward the symmetry axis before reaching a minimum beam radius downstream of the lens.

The calculated electric-field lines are also consistent with the theoretical behavior of electrostatic lenses. The field is primarily concentrated near the gaps between adjacent electrodes, whereas its magnitude rapidly decreases farther from the lens, demonstrating that the selected computational boundaries do not significantly influence the solution.

Overall, the agreement among the potential distribution, electric-field structure, particle trajectories, and beam-envelope evolution indicates that the different components of the numerical model operate consistently and provide physically realistic simulation results.

10.2 Sources of Numerical Error and Model Limitations

Like every numerical model, the present simulation is subject to several sources of numerical error and physical approximations. Although the Finite Difference Method combined with the SOR algorithm provides a stable solution of Laplace's equation, a number of factors influence the overall accuracy of the results.

The first source of error originates from spatial discretization. Replacing continuous derivatives with finite-difference approximations introduces truncation errors whose magnitude depends on the grid spacing. A finer computational grid generally improves accuracy but increases computational cost.

A second source of error is associated with the convergence of the iterative solver. If the convergence criterion is not sufficiently strict or the number of iterations is insufficient, the computed potential distribution remains slightly different from the exact numerical solution. Since the electric field is obtained from the potential gradient, these errors propagate directly into the calculated particle trajectories.

Several simplifying physical assumptions also limit the applicability of the model. Space-charge effects produced by electron-electron interactions are neglected, making the simulation

appropriate only for low-current electron beams. Furthermore, magnetic fields, collisions with residual gas molecules, relativistic corrections, and quantum mechanical effects are not included.

Finally, the electrodes are represented by idealized geometries with perfectly sharp boundaries. In practical devices, rounded edges, machining tolerances, and surface imperfections may slightly modify the electric-field distribution and consequently alter the focusing characteristics.

10.3 Possible Improvements

Although the present numerical model successfully reproduces the principal behavior of an electrostatic Einzel lens, several improvements may further increase its accuracy and computational efficiency.

One important extension is the use of adaptive or non-uniform computational grids. Such grids provide higher spatial resolution near the electrode gaps, where the electric field varies rapidly, while maintaining a coarser mesh elsewhere to reduce computational cost.

The numerical solver can also be improved by replacing the SOR algorithm with more advanced iterative methods such as Multigrid or Conjugate Gradient techniques, which generally achieve significantly faster convergence for large computational domains.

The particle-tracing module may also be extended by including space-charge effects, allowing simulations of high-current electron beams. Furthermore, higher-order integration methods such as the fourth-order Runge–Kutta algorithm could improve the accuracy of the computed particle trajectories.

Finally, the present axisymmetric model could be generalized to fully three-dimensional geometries, multi-electrode lens systems, or combined electrostatic and magnetic focusing devices, making the simulation applicable to a broader range of electron-optical systems.

10.4 Practical Applications

The numerical model developed in this work is not limited to the analysis of a single electrostatic lens and can be applied to a wide range of electron-optical systems.

One of its primary applications is the design of electrostatic lenses used in scanning electron microscopes (SEM) and transmission electron microscopes (TEM), where image quality strongly depends on accurate electron-beam focusing. Similar electrostatic focusing systems are also employed in electron-beam lithography, mass spectrometers, ion and electron sources, and particle accelerator beamlines.

Because the present simulation allows systematic modification of both geometrical parameters and electrode voltages, it provides a useful tool for optimizing lens designs before experimental fabrication. Such numerical optimization reduces development costs and shortens the overall design cycle.

Overall, the proposed computational framework provides a practical foundation for the numerical design and optimization of electrostatic electron-optical devices and can serve as the basis for more advanced simulation software in future research.

11 Optimization of Lens Parameters

The parametric study presented in the previous sections provides valuable insight into the influence of both geometrical and electrical parameters on the focusing performance of the electrostatic Einzel lens. Beyond evaluating the individual effect of each parameter, the obtained

results can also be used to identify a set of operating conditions that provides the most effective beam focusing.

From the voltage analysis, increasing the central electrode voltage was found to significantly strengthen the focusing action of the lens. Higher voltages produced shorter focal lengths and improved beam convergence by increasing the electric field intensity near the electrode gaps. Among the investigated cases, the highest applied voltage of 1200 V yielded the strongest focusing performance.

The electrode radius was found to have a considerable influence on the focusing characteristics. Smaller electrode radii generated stronger electric field gradients close to the beam axis, resulting in shorter focal lengths and better beam compression. However, excessively small radii reduce the available beam aperture and may increase the probability of beam interception by the electrode surfaces. Therefore, although a radius of 3 mm provides the strongest focusing, values around 4–5 mm represent a better compromise between focusing efficiency and practical beam transmission.

The electrode length exhibited only a weak influence on the focusing performance. Increasing the electrode length slightly modified the interaction distance between the electrons and the electric field, but the focal length and minimum beam radius remained nearly unchanged over the investigated range. Consequently, the standard length of 8 mm appears sufficient without unnecessarily increasing the overall size of the lens.

The electrode spacing also affected the focusing characteristics. Smaller gaps produced steeper potential gradients and therefore stronger focusing. Nevertheless, excessively narrow gaps may increase the risk of electrical breakdown and impose manufacturing limitations. The simulated results indicate that a gap of approximately 2 mm provides an appropriate balance between focusing strength and practical implementation.

Among all investigated parameters, the initial electron energy produced the most pronounced change in lens performance. As the electron energy increased, the focusing capability gradually deteriorated because the faster electrons experienced the electric field for a shorter time. For energies exceeding approximately 200 eV, the focal point moved beyond the computational domain, while at 400 eV and 800 eV the beam remained completely divergent. These observations indicate that the present lens configuration is optimized primarily for low-energy electron beams.

Based on the complete parametric investigation, the optimum operating conditions obtained in this study are summarized in Table 9.

Table 9: Recommended operating parameters for the electrostatic Einzel lens.

Parameter	Recommended Value
Central electrode voltage	1200 V
Electrode radius	4 – 5 mm
Electrode length	8 mm
Electrode gap	2 mm
Initial electron energy	20 – 100 eV

Overall, the optimization study demonstrates that the focusing performance of an electrostatic Einzel lens results from the combined influence of its geometrical dimensions, applied electrode voltages, and beam energy. While increasing the focusing voltage generally enhances beam convergence, practical limitations such as electrical breakdown, electrode manufacturing, and beam transmission must also be considered. Therefore, the optimum design should be selected as a compromise between focusing strength and engineering feasibility rather than by

maximizing a single parameter.

12 Conclusion

In this project, a complete numerical simulation of an electrostatic Einzel lens was developed to investigate the focusing of electron beams. The electric potential distribution inside the lens was obtained by solving Laplace's equation using the Finite Difference Method together with the Successive Over-Relaxation (SOR) iterative scheme. The resulting electric field was subsequently calculated from the potential distribution, and electron trajectories were determined through numerical integration of the equations of motion.

The numerical results demonstrated that the proposed model successfully reproduces the expected physical behavior of an electrostatic lens. The calculated potential distribution satisfies the imposed boundary conditions, while the electric field exhibits the characteristic focusing structure around the central electrode. The simulated electron trajectories clearly show beam convergence toward a focal region after passing through the lens.

The convergence history of the numerical solver confirms the stability of the iterative algorithm and its ability to obtain an accurate solution within a reasonable number of iterations. Furthermore, the calculated beam envelope provides a quantitative description of the focusing performance and allows the focal position to be estimated directly from the simulated particle trajectories.

Overall, the implemented numerical framework provides a practical tool for analyzing electrostatic lenses and investigating the influence of geometrical and electrical parameters on beam focusing. The modular structure of the developed Python code also allows future extensions without major modifications.

Possible future improvements include the incorporation of space-charge effects, relativistic particle dynamics, adaptive mesh refinement, higher-order numerical discretization schemes, and optimization algorithms for automatic lens design. In addition, the present simulation framework can be extended to more complex electrostatic or combined electrostatic–magnetic lens systems used in electron microscopy and charged-particle optics.

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